

Xraster plugin - QGIS

Raster multidimensional GIS

Climate Analysis Handbook

June 2026

Exercise Data:

All exercises in the handbook (except Exercises 4 and 5) were completed using the data from Xraster_PL_klimat.zip (1.6 GB), which can be downloaded using the link below.

Data: **https://drive.google.com/file/d/15uY5ZzUPoSzcgxppn2IjoIVe1dAKOuNw/view?usp=sharing**

(Depending on your internet speed, the download may take several minutes.)

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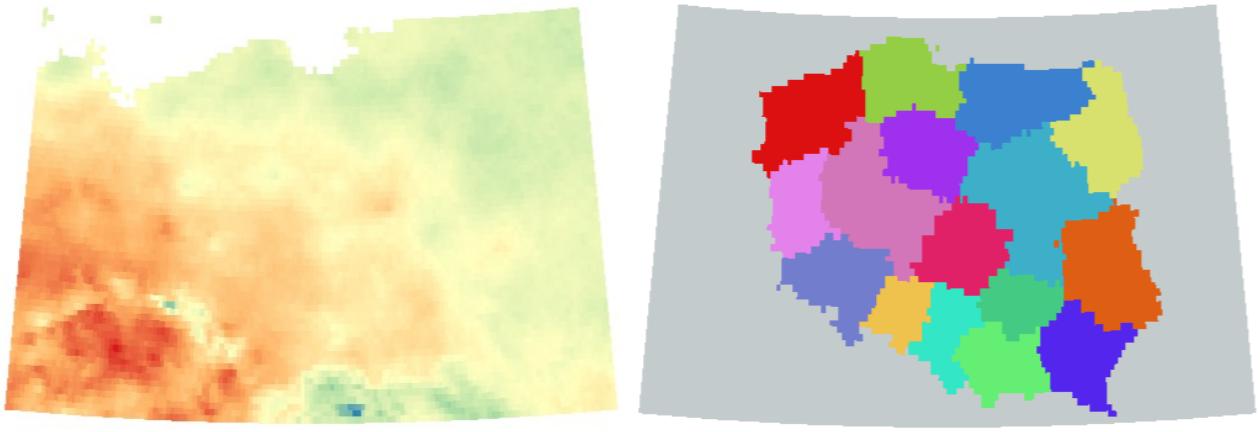
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1. Fundamentals of 3D raster data analysis (with time as a third dimension) in GIS

Raster data in GIS are usually two-dimensional data, where the dimensions are geographic coordinates (longitude, latitude) or, alternatively, projected into rectangular coordinates (X, Y) using one of the cartographic projections.

Analysis methods use two types of rasters. Continuous-value rasters (on the left below), and integer-value rasters, often treated as masks or zone maps (on the right below).



The raster with continuous values shown above contains average daily air temperatures, with NoData values over the sea. The raster with integer values contains unique (integer) values for each voivodeship of Poland. The remaining area has a value of zero. The analysis is performed on one of them (continuous or integer) or simultaneously on both the continuous and integer value maps.

Both maps cover the same area. They have a size of 62 cells by 142 cells. Originally, they are in geographic coordinates, but they are presented in the national coordinate system 1992 (PUWG 1992, EPSG:2180).

The fundamental GIS operation for raster data analysis is **map algebra** (*Raster Calculator in ArcGIS and QGIS*). It works similarly to mathematical algebra, but operations are performed cell by cell, and in the case of multiple rasters, on corresponding cells. Map algebra can perform arithmetic operations, e.g. (A - 273.15), comparisons, e.g. (A > B), or logical operations, e.g. (A > 2000) AND (B < 1000), where A and B are two different raster layers.

The second typical raster analysis operation in GIS is **reclassification** (*Reclassify in ArcGIS and Reclassify By Table in QGIS*). It involves assigning new values to existing values or their ranges. It is most often done using a table defining the assignment rules, but it can also be performed using map algebra, e.g. (A < 9)*1 + (A >= 9)*2. This assigns the value 1 to values less than nine, and 2 to values greater than or equal to nine.

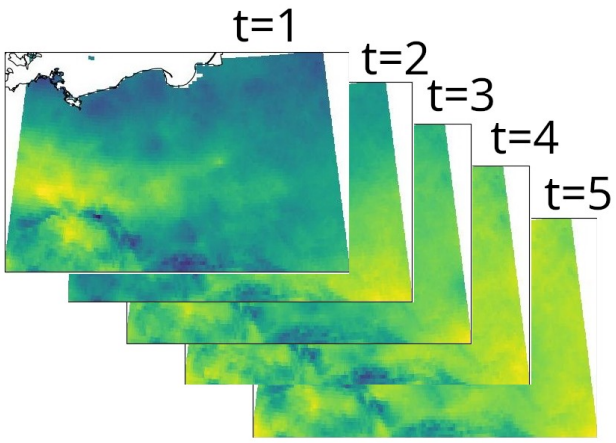
The operations described above are often referred to as **local functions** because they operate independently on raster cells or on corresponding cells across rasters.

Local functions also include functions that compute statistics from a series of rasters that are identical in spatial extent (*Cell Statistics in ArcGIS and r.series in QGIS*). The result is a raster containing the computed statistic. However, the capabilities of these functions are limited, mainly in terms of performance, to processing no more than a few hundred rasters.

Another group of raster analysis operations consists of **zonal functions**. In general, they are used to compute statistics, but also geometric parameters separately for each zone defined in a zone raster (with integer values). Examples include *Zonal Statistics As Table in ArcGIS and Zonal Statistics in QGIS*.

A further group of operations is **neighborhood functions**. As the name suggests, they operate on neighboring raster cells. They perform calculations of statistics from adjacent cells, analogous to digital image filtering operations, such as *Focal Statistics in ArcGIS and r.neighbors in QGIS*. A neighborhood operation is also the function that creates groups defined as sets of adjacent raster cells with the same value. This function assigns a unique identifier (ID) to each isolated group of cells (*Region Group in ArcGIS and r.clump in QGIS*).

In the climatic 3D raster data discussed in this handbook, the third dimension of the data is time. Each time step (e.g., a day) is represented by a separate raster. The mask or zone layer, however, remains a single layer.



In the dataset used in this handbook, each variable contains 27,394 rasters, corresponding to each day in the period 1950–2024. The analysis of such data requires a new approach, which is still largely based on an extension of the raster analysis functions described above and available in traditional GIS systems.

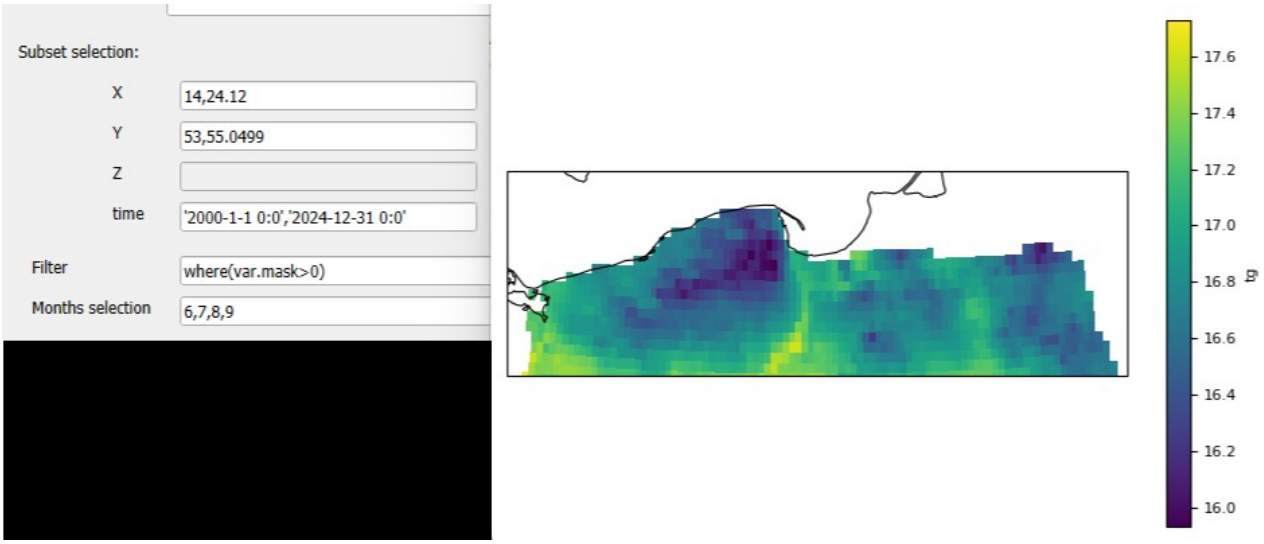
The analytical methods will focus on temporal variability of fields, and the main challenge will be the number of rasters that must be analyzed simultaneously.

New concepts defining these methods will be introduced.

Subset (or slice) of data,

contains a single variable from the dataset, which may be constrained in both the spatial and temporal dimensions. In the spatial dimension, it may be limited by coordinates or a mask, and in the temporal dimension by a defined time period or selected months.

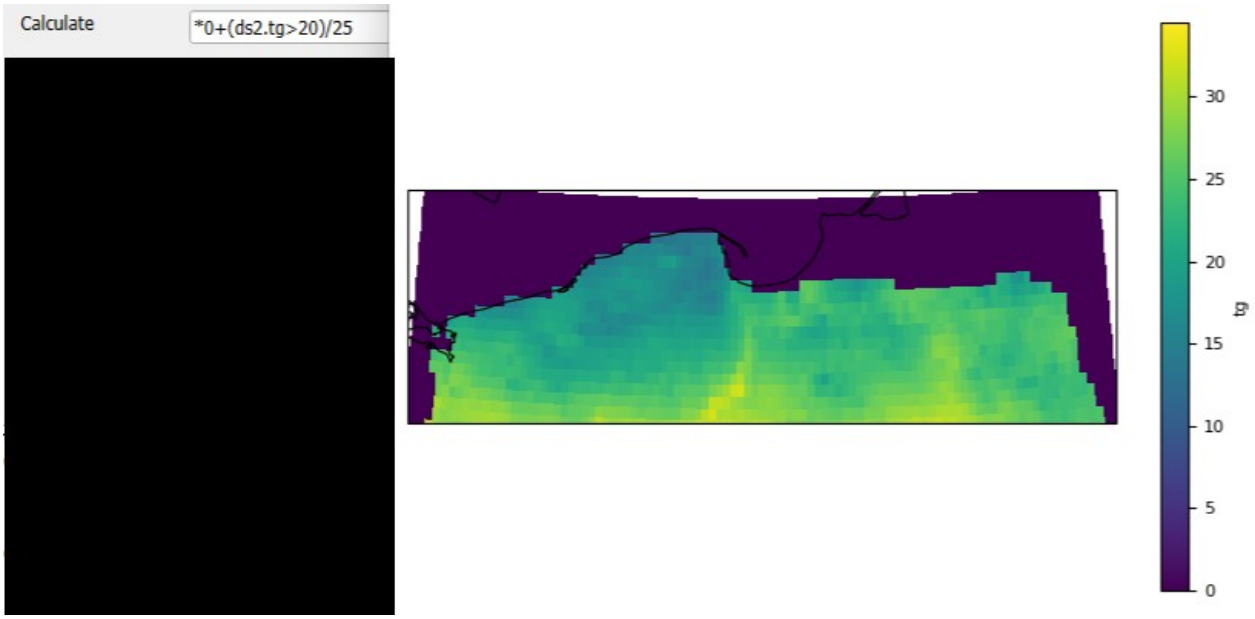
The figure shows the definition of a subset (X, Y, time, mask > 0 – only the area of Poland, and a restriction of the subset to the months from June to September) of mean daily air temperature. From such a defined subset, a layer of average values was calculated.



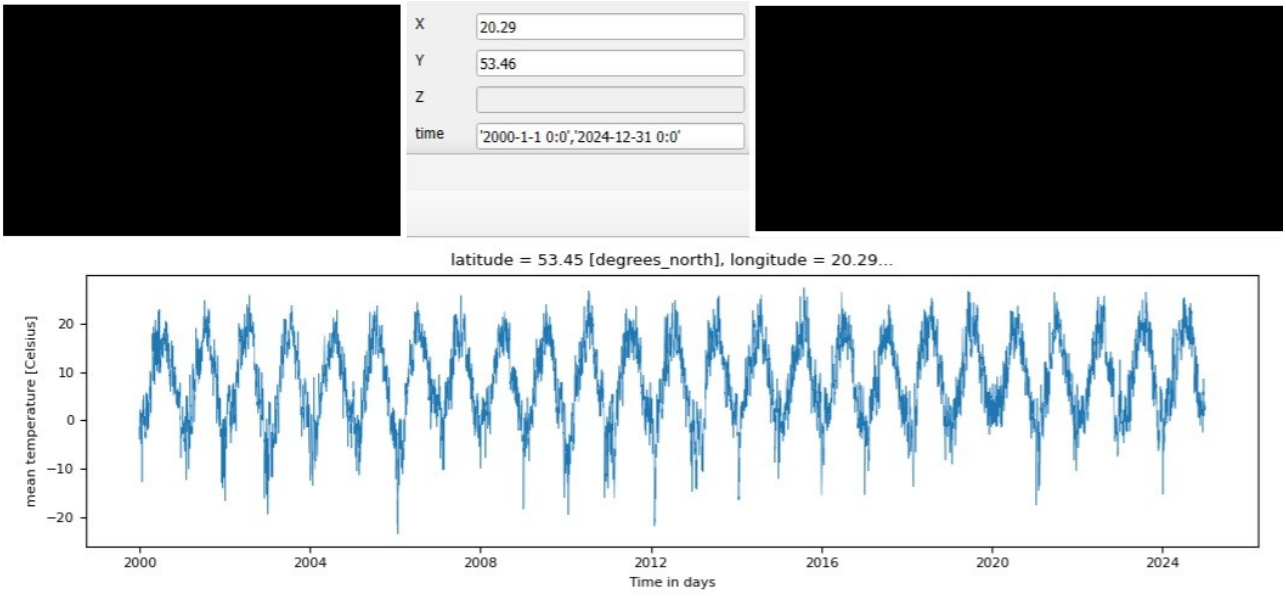
Creating a subset is the first and fundamental step of the analysis.

A variable in a subset may be the result of map algebra (Calculate).

The same subset was converted to values of 1 (when the temperature > 20) and 0. Then, the corresponding cells were summed. The sum was divided by the number of years to determine how many days per year, on average, a given cell has a temperature greater than 20 degrees. The method for creating expressions will be discussed during the exercises.



In the case of limiting the subset to a single point, the result is a time series.



Anomalies.

In climatology, anomaly analysis plays an important role. Variables in a subset can be converted into variable anomalies. A spatial anomaly refers to the difference between the observed value of a given variable at a specific geographic location and its long-term mean (climatological mean) for the same place. It shows how much the conditions at a given point in space (e.g., in a data grid cell) deviate from what is usually observed there.

Anomalies can be calculated using the expression:

$A(x,y,t) = X(x,y,t) - X_{mean}(x,y)$

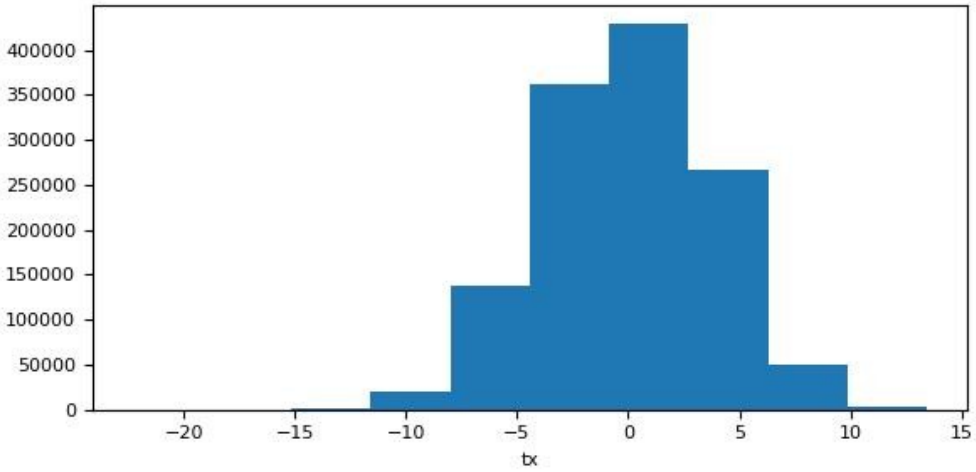
where:

$X(x,y,t)$ - the observed value of the variable at location (x, y) and time t

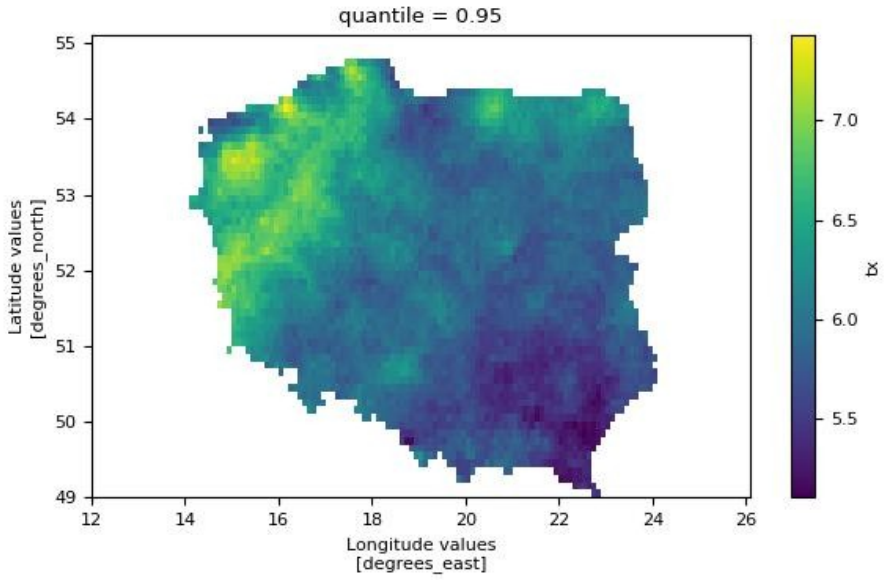
$X_{average}(x,y)$ - the mean at that location

$A(x,y,t)$ - anomaly.

For the summer months (July, August) in the years 2020–24, such defined anomalies were calculated for the area of Poland for daily maximum temperatures. Their histogram has the following form,



The range of most anomalies is from -5 to +5 degrees, with a distribution close to normal. From such a transformed subset, we can create a map of statistics. The 0.95 quantile (or 0.05) is often used to show a map of anomalies for which there are more or fewer than 5%. The map created in this way is shown below.



This type of anomaly (sometimes called seasonal) applies to periods without clear cyclical variability. The mean will be calculated from 5 (years) × 62 (summer days) at each location.

The second classic method of calculating anomalies is to compute them based on the mean separately for each day of the year, most often over a 30-year period. The expression will take the form,

$$A(x,y,d,t) = X(x,y,d,t) - X_{mean}(x,y,d)$$

where:

X(x,y,d,t) - observed value of the variable at location (x,y) on day d of year t

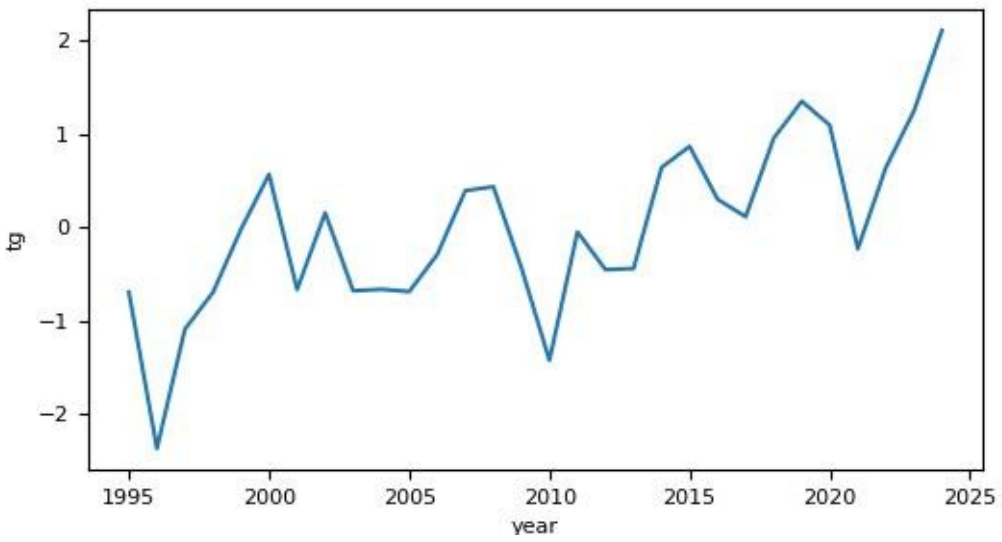
$X_{mean}(x,y)$ - long-term mean at this location for day d of the year

A(x,y,d,t) - anomaly for day i and year

d - day of the year (1 -365 or 366)

t- year of observation

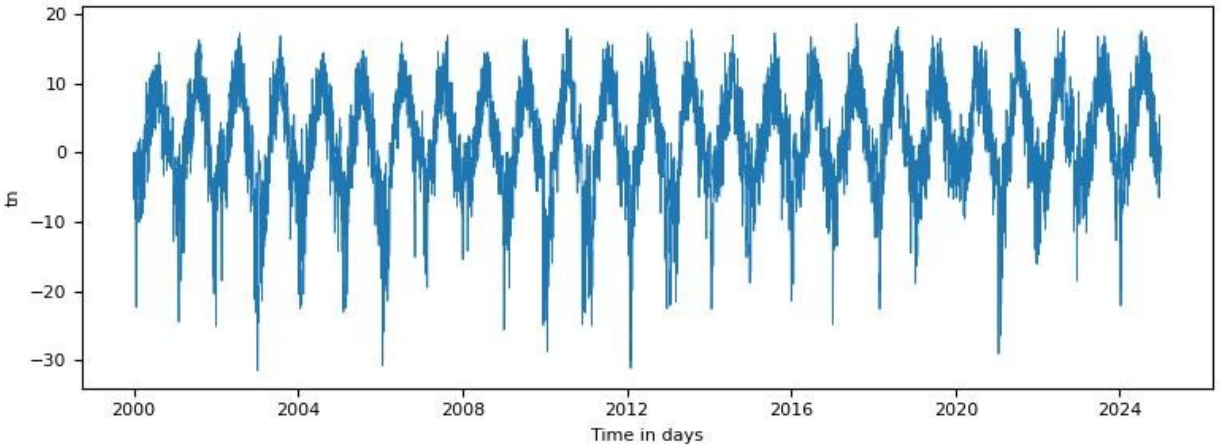
In this case, at each location the mean is calculated from 30 values. The figure below shows the time series of annual mean anomalies of daily mean temperature (from the area of Poland) over the last 30 years (1995–2024).



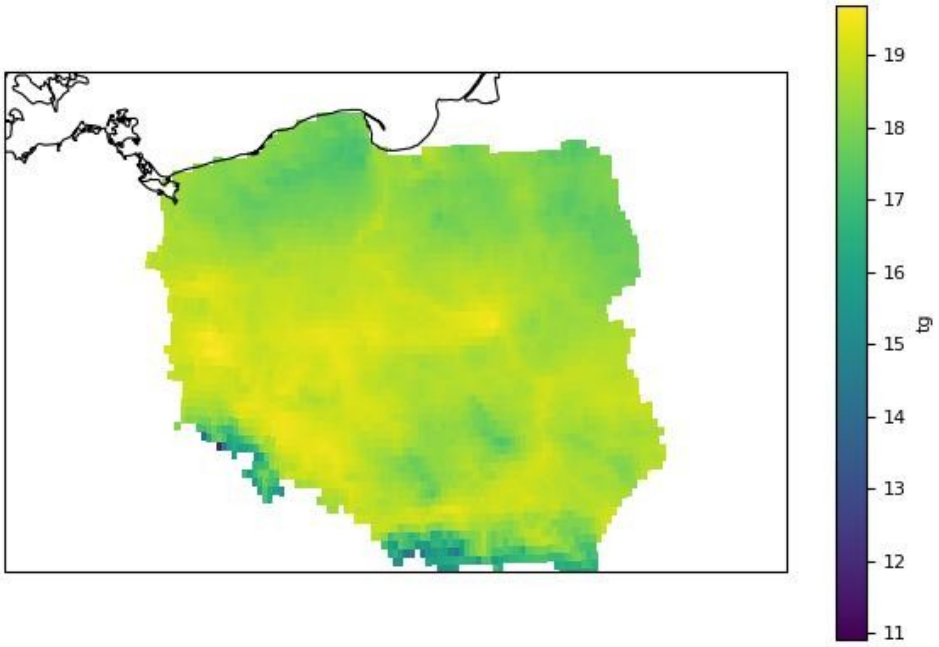
Statistics.

The two basic types of statistics are **spatial** and **temporal summarization**.

Spatial summarization calculates statistics separately for each moment in time over the entire area or defined zones. It therefore performs the functions of zonal statistics. The results are saved in a text file separately for each zone for each moment in time, or in the form of a time series plot for the entire subset. The chart above shows daily minimum temperatures in a selected province over the period 2000–2024.

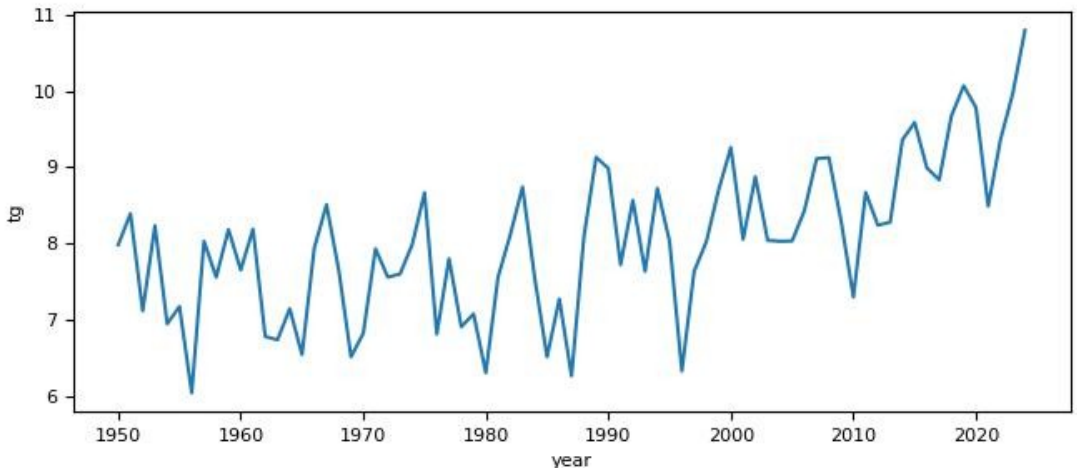


Temporal summarization (time summary) calculates statistics for each raster cell over the analyzed time period. It therefore works like *Cell Statistics*. The figure below shows a map of mean August temperature for the area of Poland over the last 30 years.

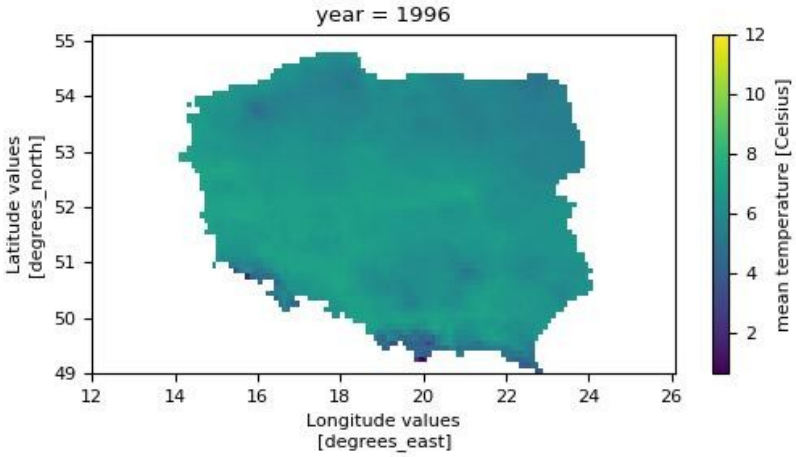


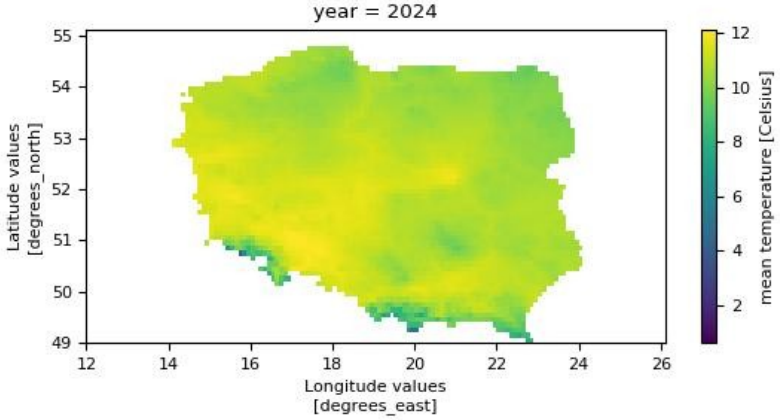
Group statistics (over time).

Grouping consists of dividing the data into groups defined by time dimensions (years, months, days, climatic seasons). Statistics are then calculated separately for each group and presented on a single chart for analytical purposes. At the same time, maps with statistics for each group are created. The figures below show changes in the average annual air temperature for the area of Poland.



Below are maps showing the average temperature for years with extreme values.

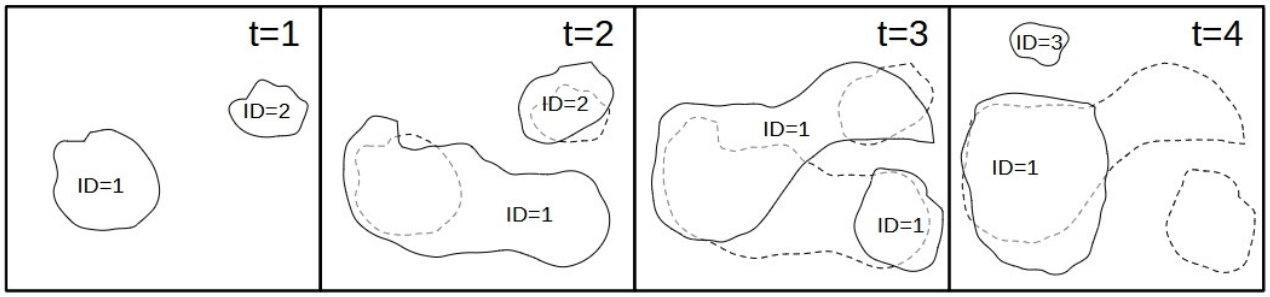




Spatio-temporal groups (regions) and their analysis.

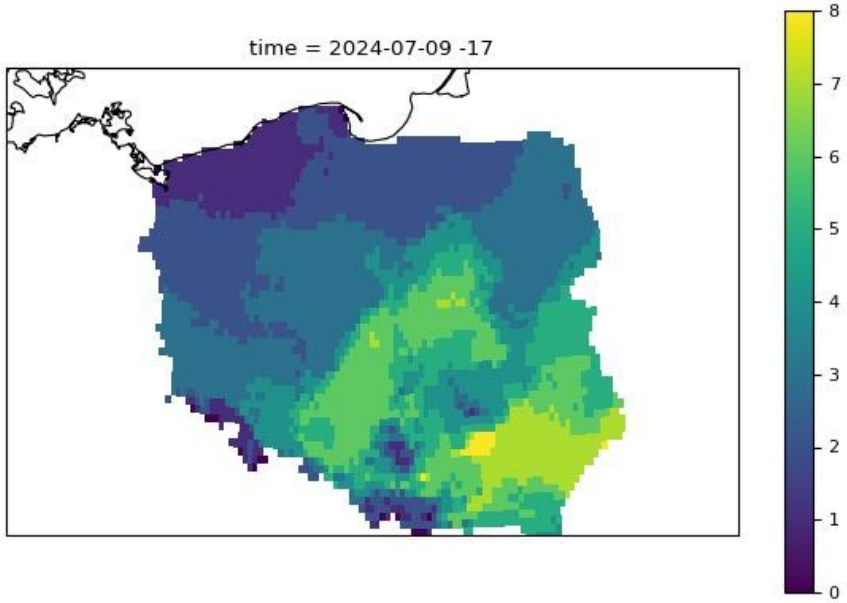
Spatio-temporal groups (or regions) are an extension of the concept of region groups from two-dimensional raster space to three-dimensional space that additionally includes time.
ST region groups are created based on the following principles:

1. Groups maintain continuity both in space and time.
2. Each group has a unique ID.
3. A group retains its ID in the next time step if its new position partially overlaps with the previous one.
4. Merged groups with different IDs adopt the ID of the group that contributed the largest area.
5. A group may split, creating a multi-element group (the separate groups retain the same ID).

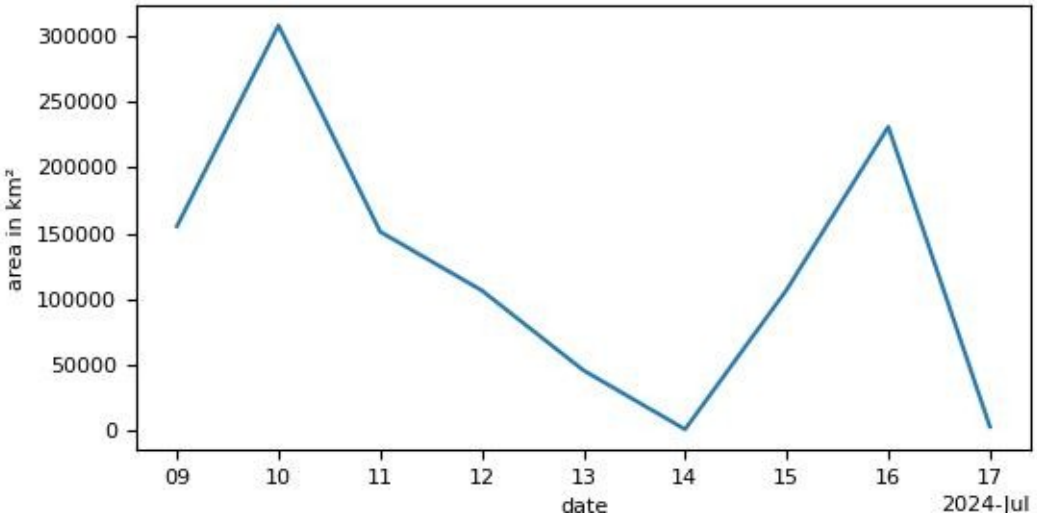


The process of defining spatio-temporal region groups is presented above. Subsequent time steps are labeled as t=1, 2, 3, 4. The dashed line indicates the position of groups in the previous time step. In the first time step (t=1), groups are defined solely based on continuity.

An example application of this method is the identification of heatwaves, which can be defined as a sequence of days (for example, more than 3 days) with a maximum temperature above 30 degrees over a spatially continuous area. This analysis allows for the identification of individual groups (heatwaves) and the creation of maps showing the number of days for the defined group.



The map shows a heatwave that occurred from July 9 to July 17, 2024. The map includes the duration of the heatwave in days. A given group (heatwave) can also be represented on a chart showing how its spatial extent changed over time.



2. Climate data in the Copernicus Climate Data Store

Climate data are most commonly provided in one of two formats: NetCDF and GRIB. Currently, Xraster works exclusively with data in **NetCDF** format.

NetCDF is a binary format designed for storing multidimensional scientific data along with descriptive metadata. It allows multiple variables (data) to be stored as multidimensional arrays, e.g. **time × latitude × longitude × altitude**. This makes it well suited for storing meteorological and climate fields.

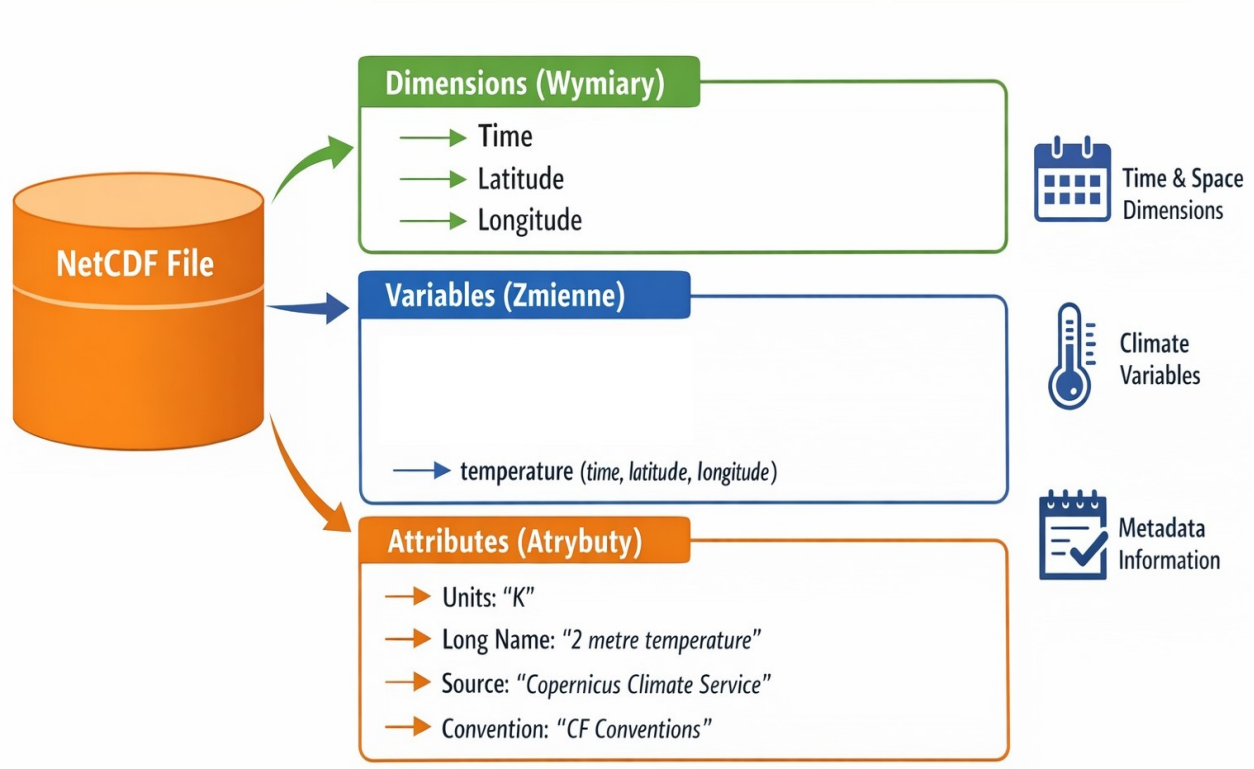
A typical **NetCDF** file with climate data may contain:

- **a time dimension** – e.g. hours, days, or months,
- **spatial dimensions** – latitude and longitude of the grid,
- **climate variables** – e.g. surface temperature, wind speed, precipitation.

Thanks to this structure, it is easy to extract data for a specific region or time period.

Below is a schematic diagram of a NetCDF file.

Structure of a NetCDF file



The basic dataset that can be used in climate analyses of European countries is the ***E-OBS daily gridded meteorological data for Europe*** from 1950 to the present, derived from in-situ observations.

E-OBS is a European observational climate dataset with daily resolution, covering the land areas of Europe. The data are presented on a regular spatial grid and are produced based on measurements from weather stations collected within the European Climate Assessment & Dataset project.

Key facts

- Data source: a network of meteorological stations provided mainly by European National Meteorological and Hydrological Services (NMHS) and other institutions collecting climate data.
- Spatial coverage: land areas of Europe.
- Temporal resolution: daily data (one value per 24-hour observation period).
- Data format: regular spatial grid (gridded dataset).
- Climate variables: the dataset includes several basic meteorological parameters, including temperature, precipitation, and other climate indicators.

Distinguishing features

- high density of meteorological stations in many countries, often covering entire national measurement networks,
- long time series of data,
- daily resolution and multiple climate variables

The E-OBS dataset contains several basic meteorological variables describing daily climatic conditions over the land areas of Europe. The data mainly refer to measurements taken near the Earth's surface (usually at about 2 m height).

The dataset includes the following variables:

- Maximum air temperature – daily maximum near-surface air temperature (usually at ~2 m above ground), expressed in °C.
- Mean air temperature – daily average near-surface air temperature (~2 m), in °C.
- Minimum air temperature – lowest air temperature recorded during the day, in °C.
- Precipitation total – total daily precipitation (rain, snow, or hail) expressed in millimetres as water equivalent. The data come from rain gauges, although the exact 24-hour observation period may vary between countries.
- Relative humidity – daily mean near-surface air humidity (~2 m), expressed in percent. Values range from 0 to 100%, where 0% indicates completely dry air and 100% indicates full saturation with water vapour.
- Sea level pressure – mean daily atmospheric pressure reduced to sea level, expressed in hectopascals (hPa).
- Surface shortwave radiation – flux of solar radiation reaching the Earth's surface, expressed in W/m².
- Wind speed – mean daily wind speed measured at 10 m above the ground, in m/s.

For the purposes of this handbook and climate data analyses for Poland, a subset of these data has been prepared for the area of Poland: **PLCL_obs_1950_2024.nc**, containing a range of variables. Additionally, a voivodeship mask **PLmask_woj.tif** has been created for this dataset, which can be modified in QGIS.

The data files (dataset and mask) can be downloaded from:
Download dataset and mask

Below is the dataset information for the **PLCL_obs_1950_2024.nc** dataset:

Dimensions:
time, 27394
latitude, 62
longitude, 142

Coordinates:
longitude, 11.9499, 26.0499
latitude, 48.9499, 55.0499
time, 1950-01-01 00:00:00, 2024-12-31 00:00:00

Variables:
tg [°C] mean temperature
tn [°C] minimum temperature
tx [°C] maximum temperature
rr [mm] rainfall
hu [%] mean relative humidity
qq [W/m²] surface downwelling shortwave flux in air

This dataset was created using **Xraster** from original data downloaded from the **Copernicus Data Store** using Clip/Merge tools.

Daily meteorological data for the entire globe are contained in the dataset **ERA5 post-processed daily statistics on single levels from 1940 to present**.

ERA5 is the fifth generation of climate and weather reanalysis developed by the **European Centre for Medium-Range Weather Forecasts (ECMWF)**. It provides global data covering the last ~80 years, available from 1940 onward, and replaces the earlier ERA-Interim reanalysis project.

Reanalysis is the process of combining numerical model outputs with observational data from around the world into a single, consistent global dataset. This process uses the laws of physics and a method called **data assimilation**. The method works similarly to numerical weather forecasting: every certain period (at ECMWF, every 12 hours), a previous forecast is combined with new observations. As a result, the best estimate of the current state of the atmosphere is produced, which is then used to generate an improved forecast.

In the case of reanalysis, the process is similar, but a lower computational resolution is used in order to create a long-term dataset spanning many decades. Reanalysis does not need to provide real-time forecasts, so more observations can be collected, including improved versions of previously processed data. This results in higher quality and more temporally consistent datasets.

ERA5 data description (summary)

- **Data type:** gridded data
- **Projection:** regular latitude–longitude grid

- **Spatial coverage:** global

Horizontal resolution:

- Reanalysis: 0.25° × 0.25° (atmosphere), 0.5° × 0.5° (wave ocean)
- Ensemble members: 0.5° × 0.5° (atmosphere), 1° × 1° (wave ocean)
- **Temporal coverage:** from 1940 to present
- **Temporal resolution:** daily
- **File format:** NetCDF (in ZIP archives)

Version: currently only one product version is available; if changes in processing occur, versioning will be introduced (this version will be labeled v1.0).

Data source note: daily statistics are computed from hourly ERA5 data; differences between ERA5 and ERA5T should be taken into account.

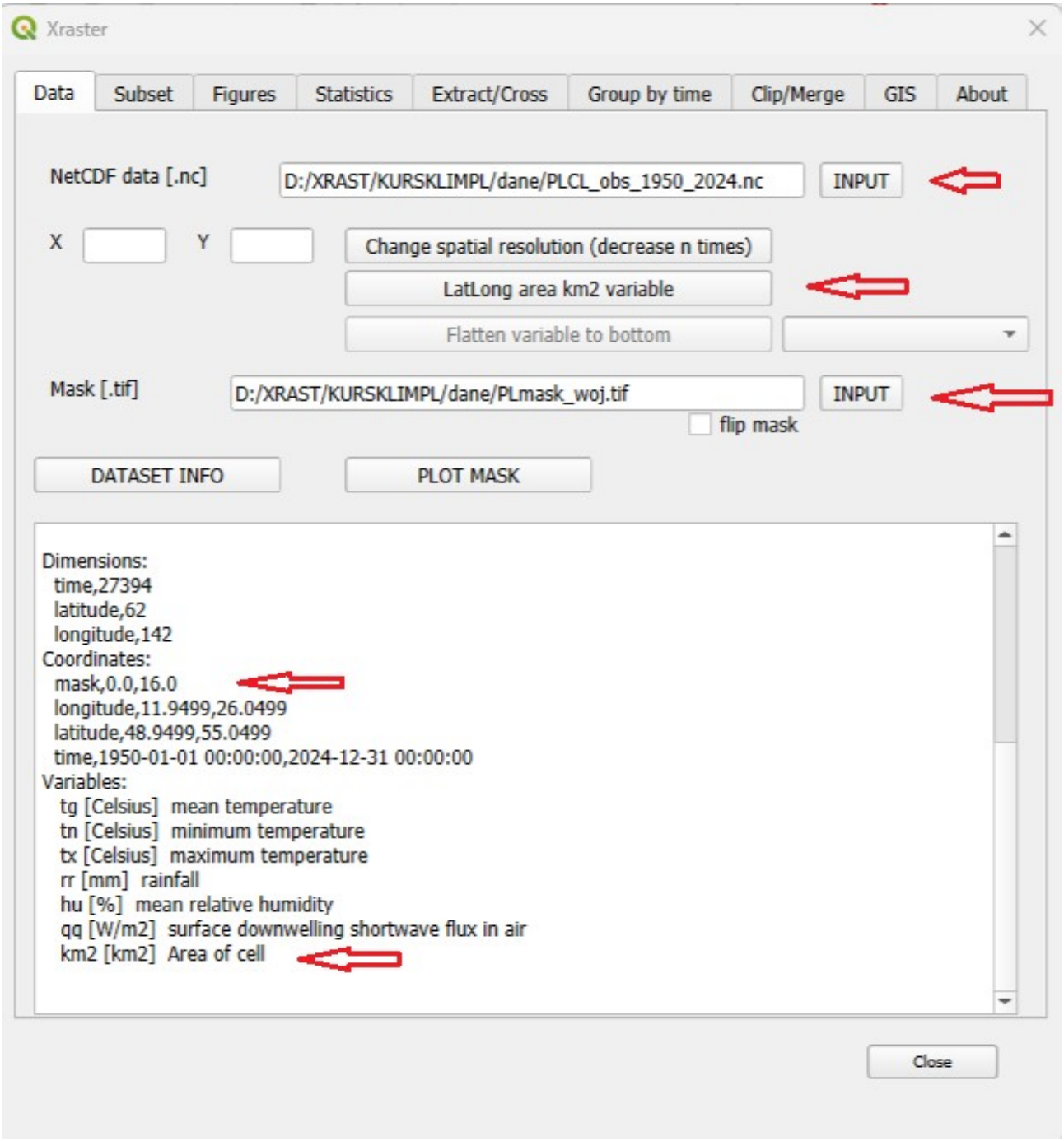
Update frequency: daily, with an approximately 6-day delay.

This dataset contains a very large number of variables.

3. Example climate analyses using Xraster.

The analyses will be conducted using answers to subsequent questions. For the initial questions, we will limit the area to northern Poland.

The first step will be to **input the data and the administrative units (voivodeships) mask**. We will also create an additional variable containing the area of the raster cells.



We open Xraster. In the Data tab, we perform three operations. We load the .nc data, load the mask, and add a variable with cell areas in km² by clicking the **LatLong area km2 variable** button. The **DATASET INFO** button allows us to refresh the dataset information.

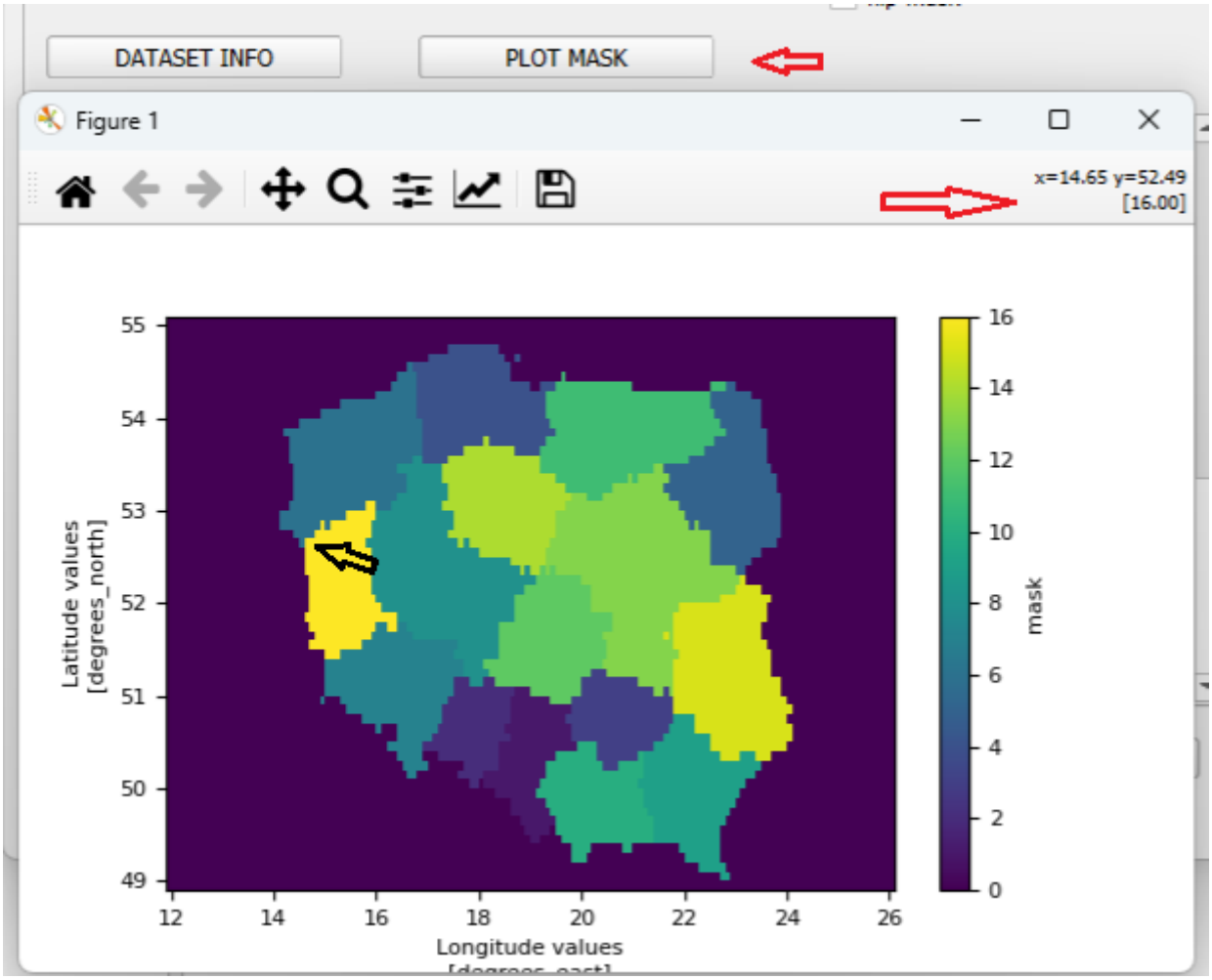
Dimensions describes the spatial size of the raster. For each variable, we have 27,394 rasters with a size of 62 × 142 cells.

Coordinates provides information about the ranges of dimensions. **Time** shows that the (daily) data span from the beginning of 1950 to the end of 2024. Additionally, in **Coordinates**, we have information about the values contained in the mask (0 to 16).

Variables describes the variables contained in the dataset. We have six variables describing climate and one variable describing area.

In practice, the dataset contains 27,394 × 7 = 191,578 rasters plus the mask.

Using the **PLOT MASK** button, we can display the mask.

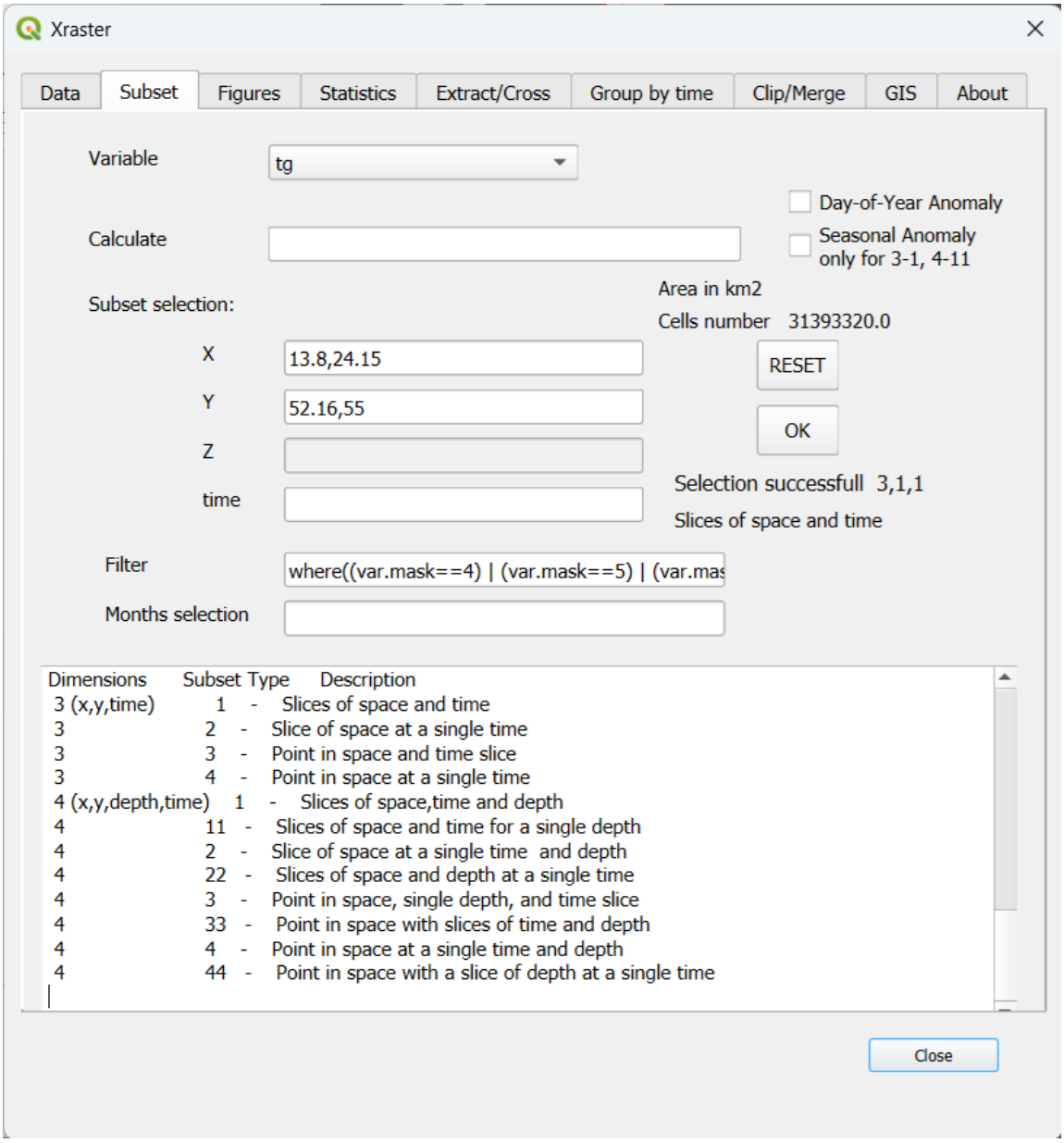


By moving the cursor, we can read the values on the mask (the northern voivodeships have identifiers 4, 5, 6, and 11, and are covered by the latitude range 52.16–55 and longitude range 13.8–24.15).

We will limit our analysis to the northern voivodeships and also restrict the map to the latitude and longitude ranges specified above. To create the subset, we will use a filter of the form:

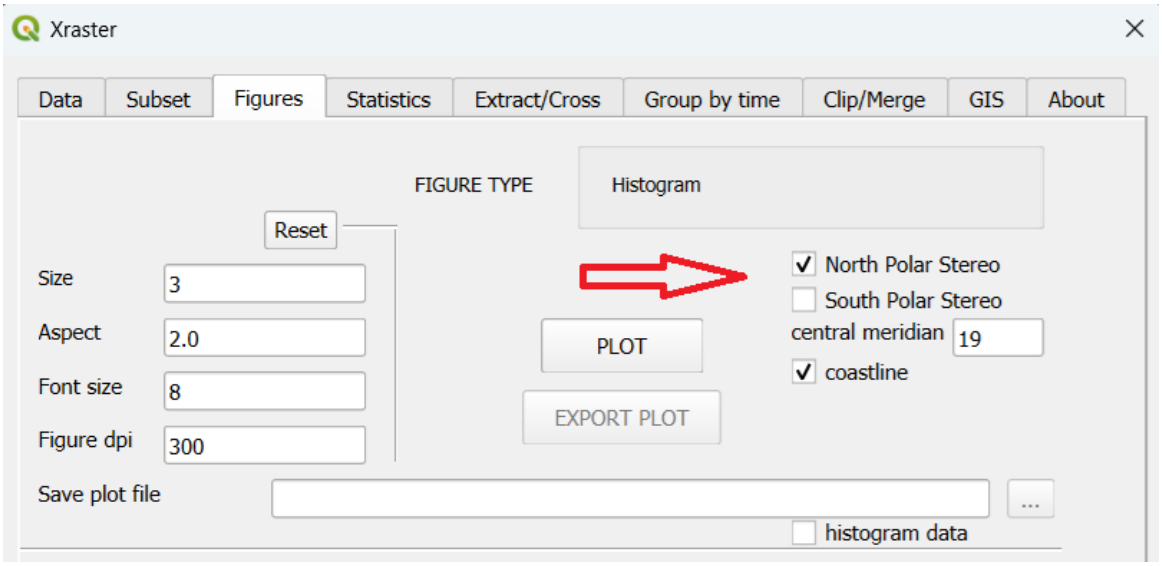
```
where((var.mask==4) | (var.mask==5) | (var.mask==6) | (var.mask==11)).
```

The subset for the mean temperature variable will have the following form,



After entering the parameters, we click the **OK** button. Information about the type of subset created will appear below the button.

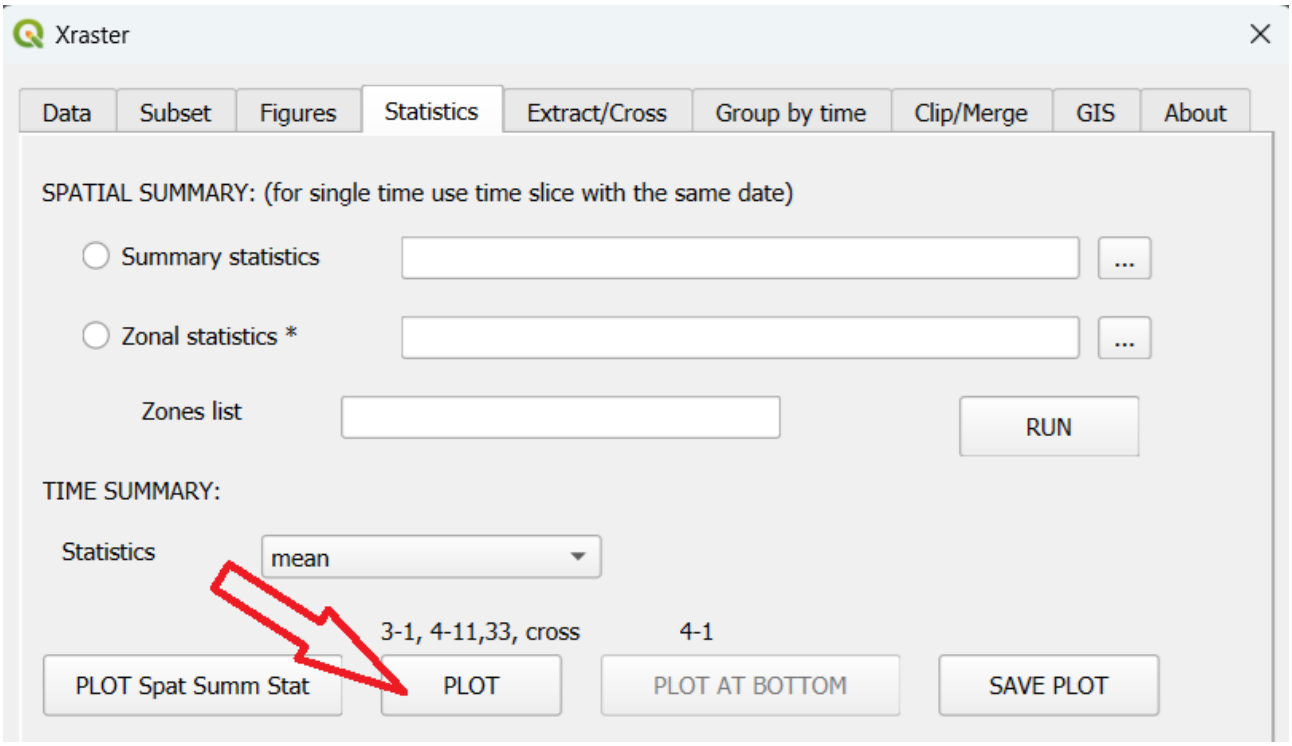
Additionally, in the **Figures** tab, we will set the options so that the maps are displayed in a projected coordinate system with a coastline.



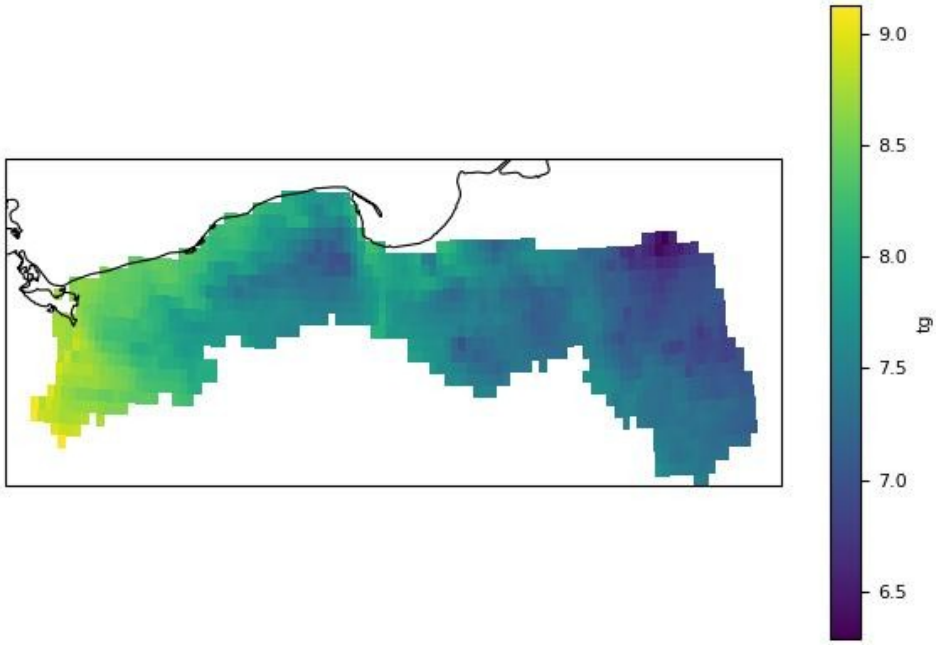
The analysis will be conducted by asking a series of questions.

What the average annual temperature map of northern Poland looks like (for the period 1950–2024)

To obtain a map of mean values for the entire time period, we go to the **Statistics** tab, select the **mean** statistic, and click the **PLOT** button.

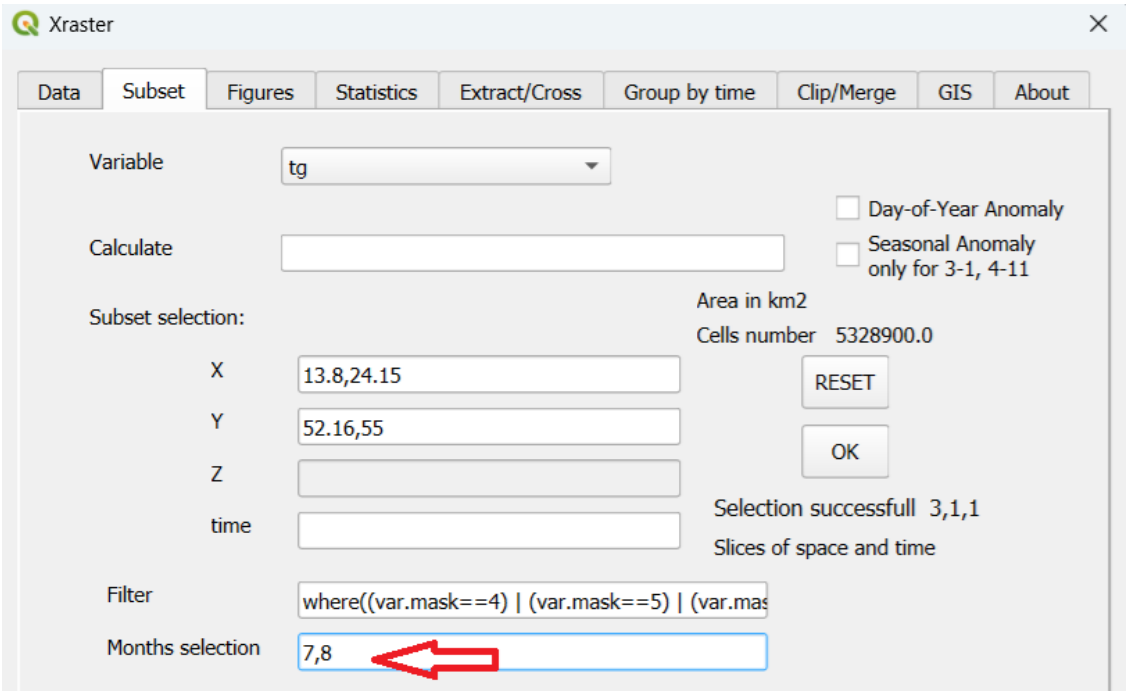


We will obtain a map of the following form,



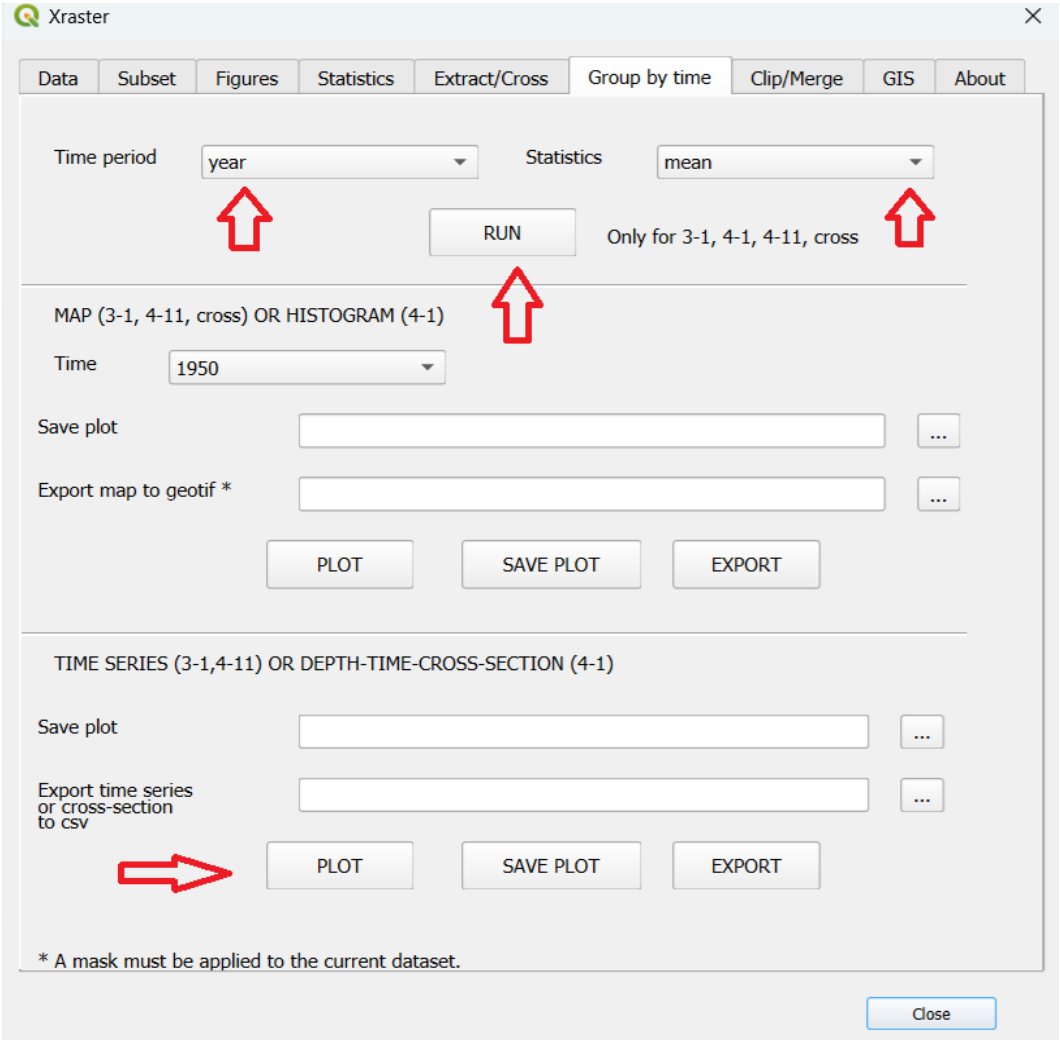
Comparison of average temperature maps for the coldest and warmest summers (July, August) in northern Poland (for the period 1950–2024)

To obtain a subset only for the summer months, we add a month restriction when creating the subset in the form,



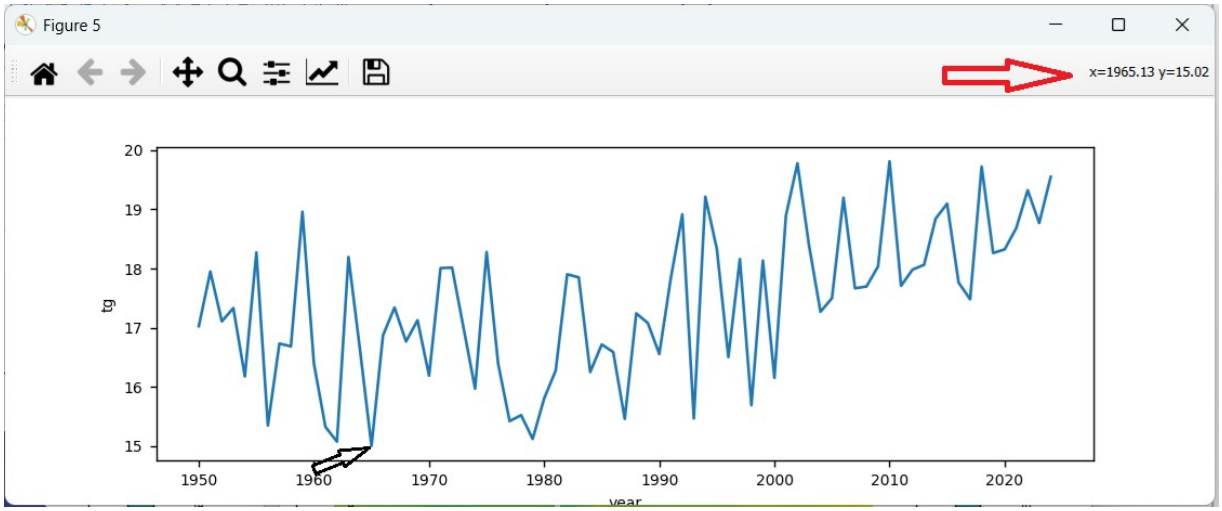
We now create a new subset by clicking **OK**.

We will now determine the years with the minimum and maximum temperature in July and August. We go to the **Group by time** tab and group the data by years (**Time period**) using the **mean** statistic.



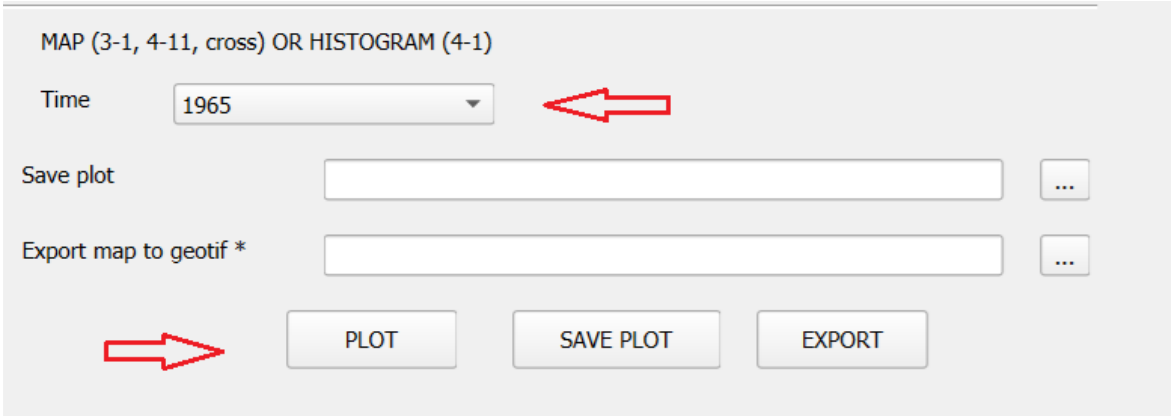
We click **RUN** and **PLOT (in TIME SERIES...)**.

A time series is displayed, in which years with the minimum and maximum mean values can be identified.



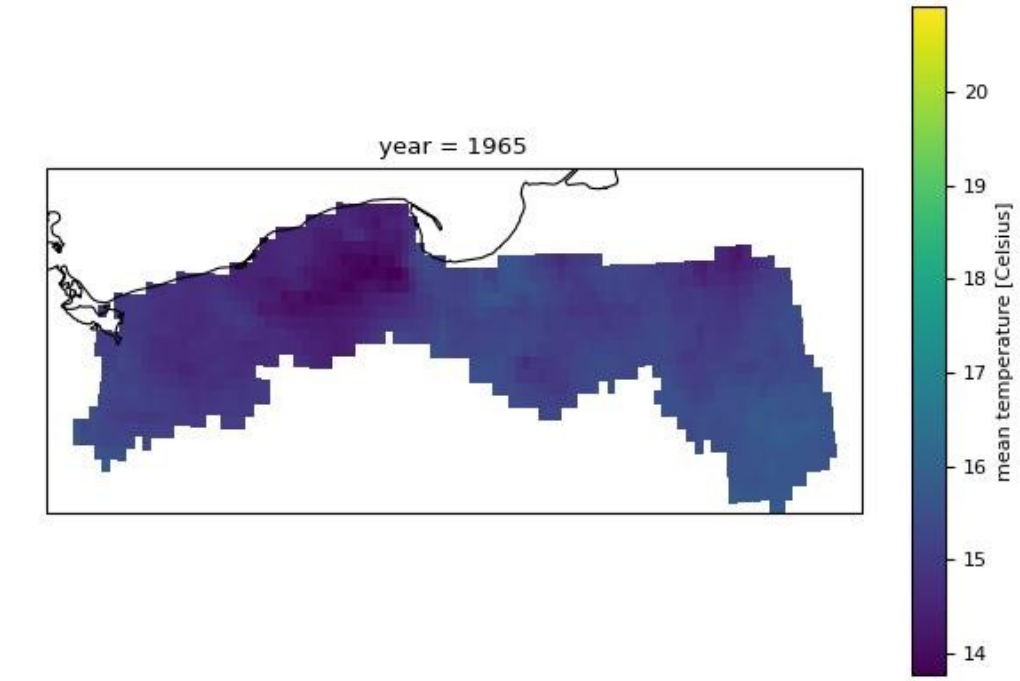
We can see that the difference in mean temperature was approximately 5 degrees. A clear increasing trend in average summer temperature after the year 2000 is also visible.

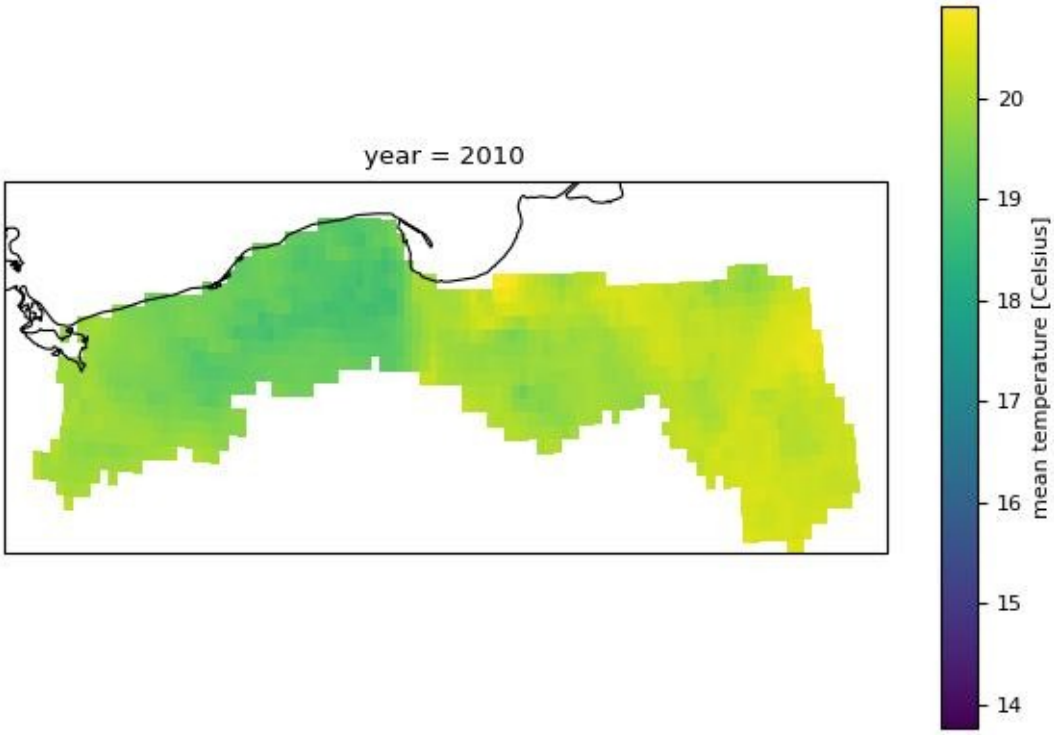
Maps for these years can be created in the central part of the tab for the selected years 1965 and 2010.



These maps can be converted to a common color palette for better comparison.

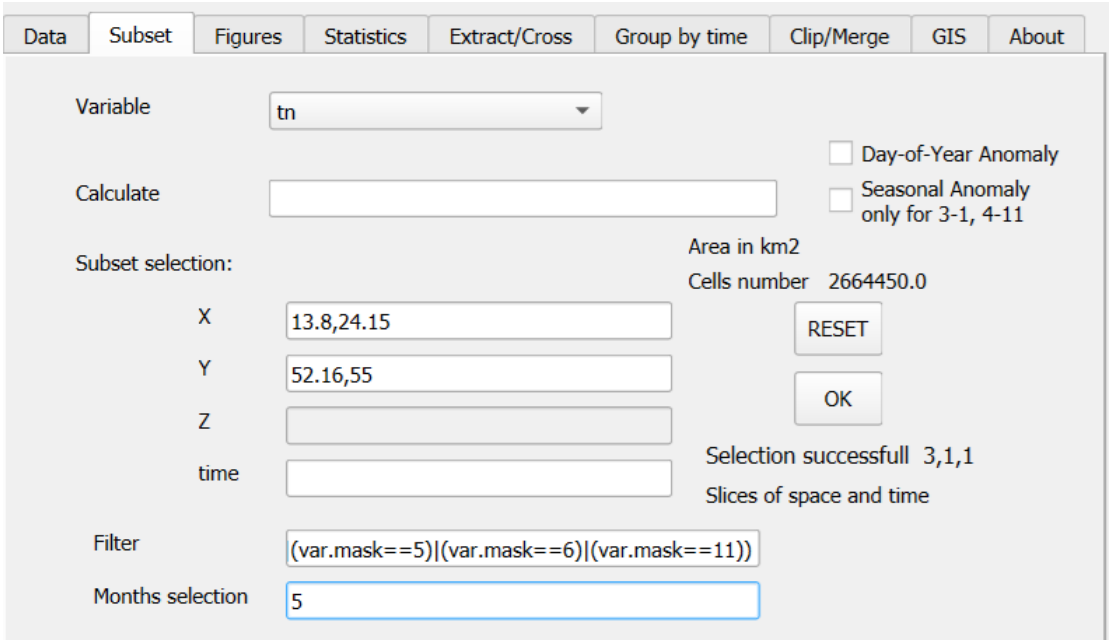
The results are presented below.


year = 1965
20
19
18
17
16
15
14
mean temperature [Celsius]

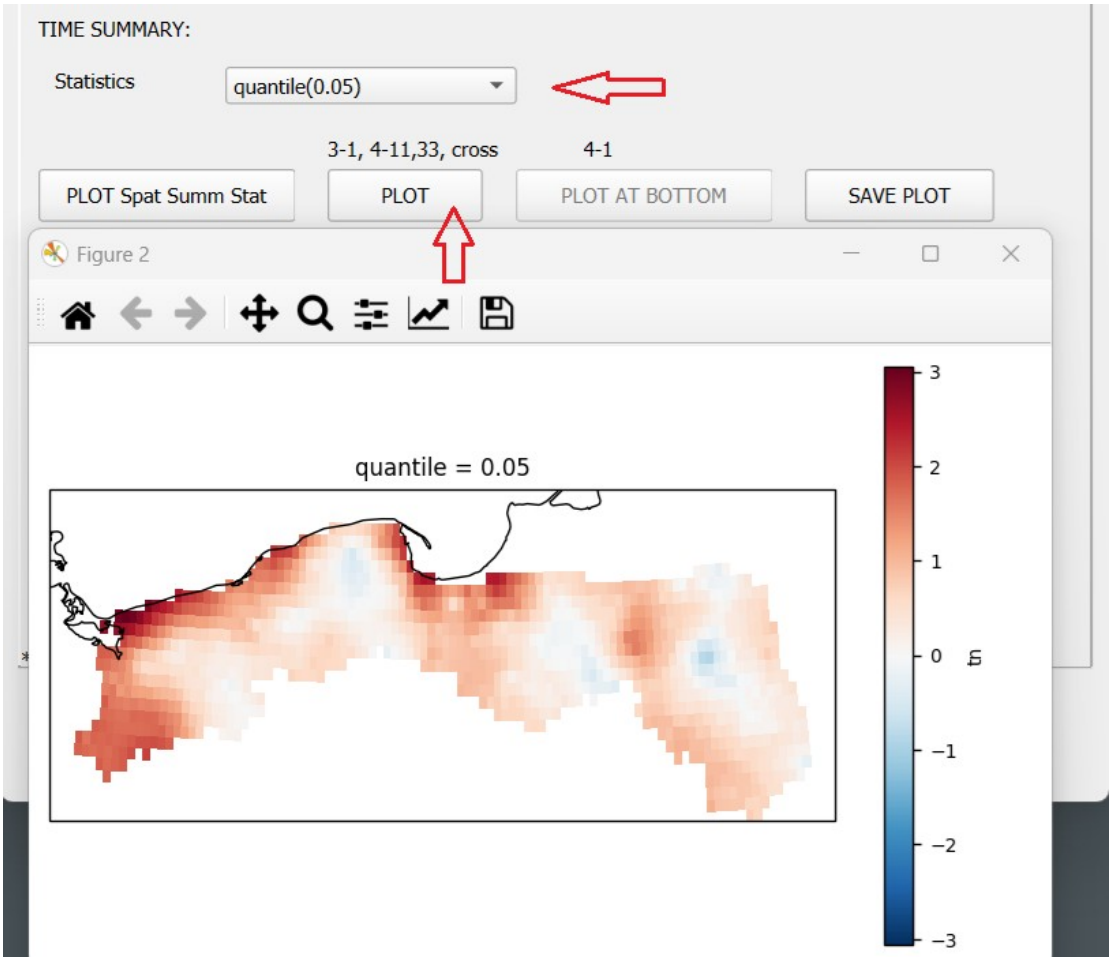

year = 2010
20
19
18
17
16
15
14
mean temperature [Celsius]

Distribution of frost events (minimum temperature below 0°C) in May in northern Poland (for the period 1950–2024)

To obtain a subset for the minimum temperature **tn** only for May, we modify the subset as follows.

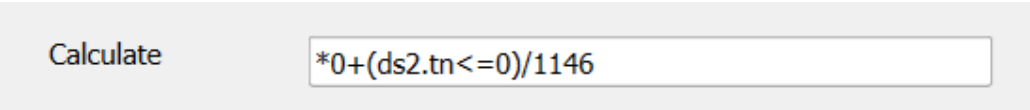


To visualize the distribution of frost events, we will use the 0.05 quantile (the value above which 95% of the minimum temperature values are higher).



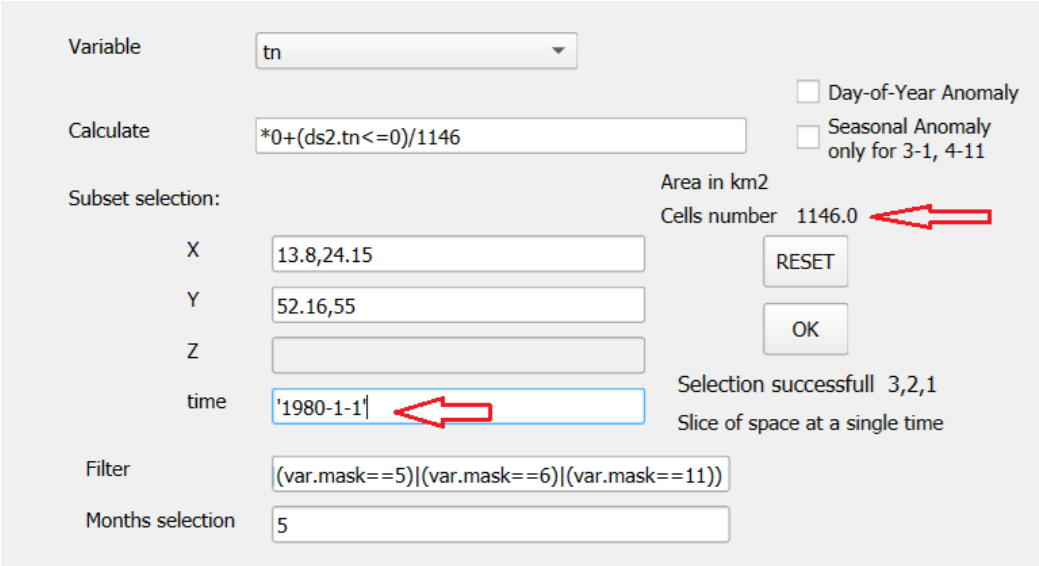
We can see that there are several centers of frost occurrence. We will examine whether there is a trend in the number of frost days across the entire area and identify which year was record-breaking in terms of frost events.

To carry out this analysis, we need to assign a value of 1 to a raster cell if frost occurred on a given day, and 0 if it did not. We do this using the following expression:

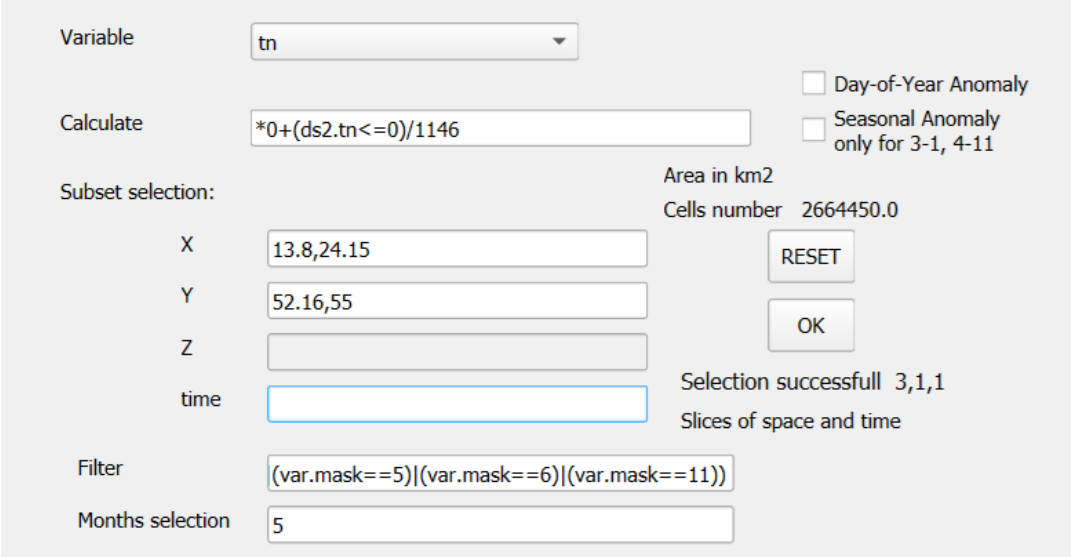


First, we set the temperature values to zero (*0), then add the result of the logical expression (ds2.tn <= 0), which returns either 1 or 0. The notation *ds2.tn* is a generic way of referring to any variable.

Since we want to plot the average number of frost days per year across the study area, we calculate the total number of frost days and divide it by the number of cells in the area. The number of cells in the study area can be obtained by performing any operation on a single day's data.

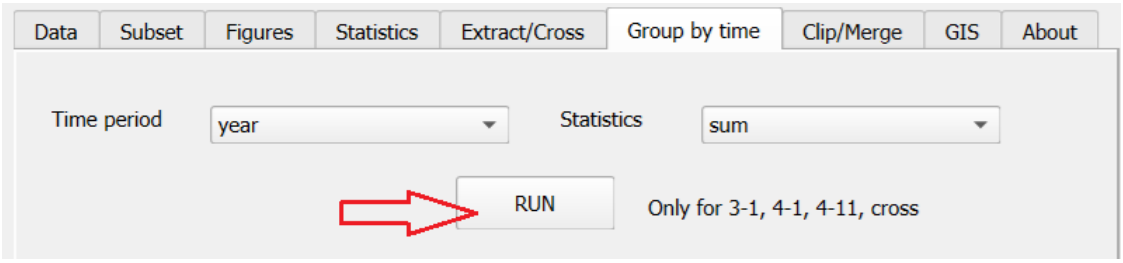


We now apply this subset to the entire time period.

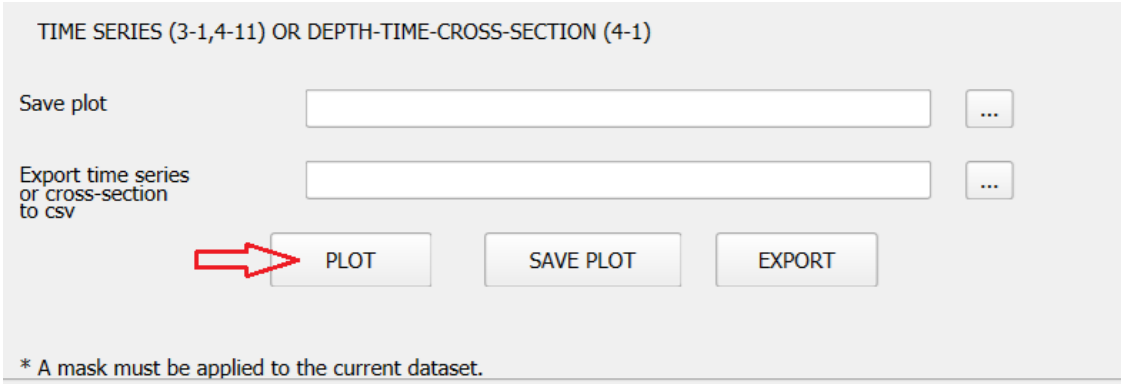


Then we go to the **Group by time** tab, set the *Time period* to year and *Statistics* to sum. The program will then, for each year (May), sum all values (1 or 0) across all cells.

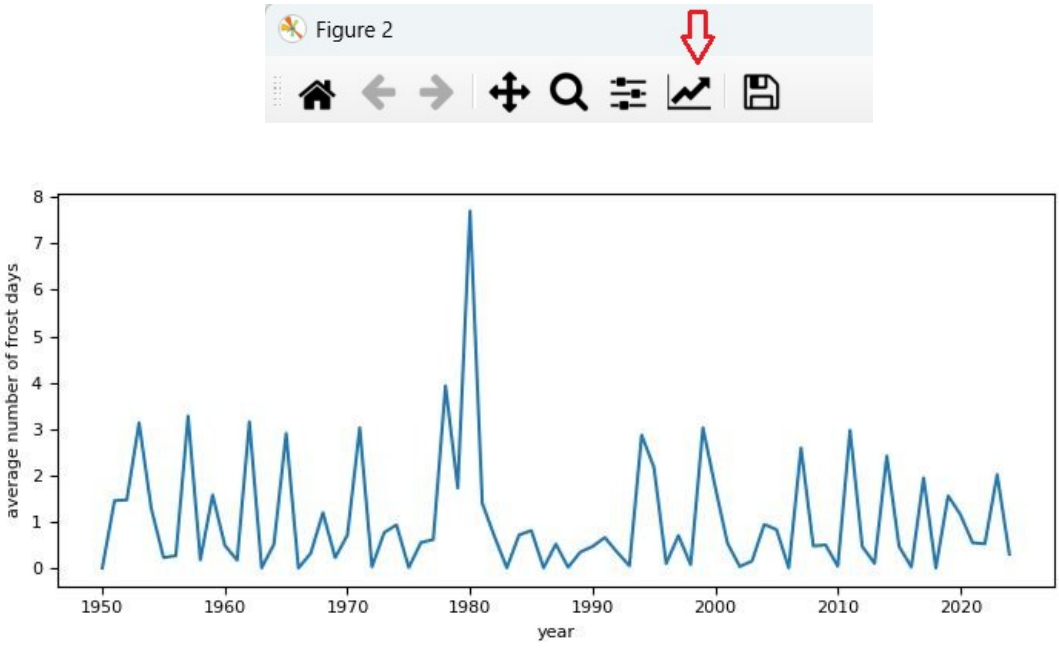
If on a given day there were frost conditions in the whole area, and on all other days no frost occurred anywhere, the total sum for the entire area would be 1146. After dividing by the number of cells, this would give an average value of 1.



At the bottom of the panel, we click “PLOT”.

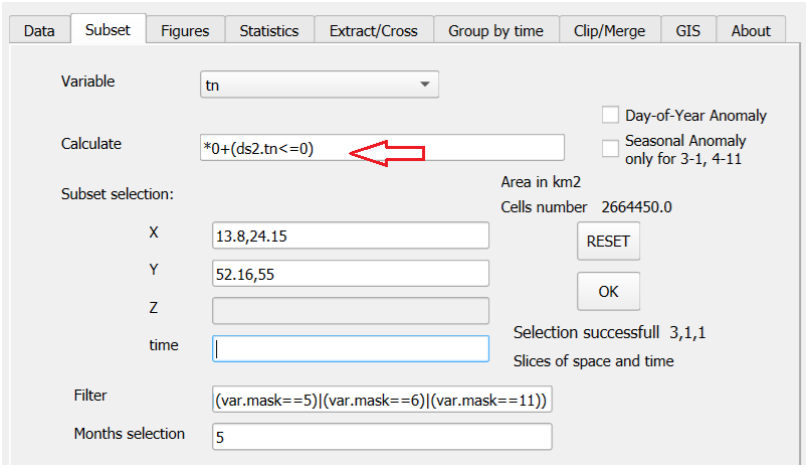


We obtain a time series plot of the average number of frost days (the axis labels have been changed in the figure properties).

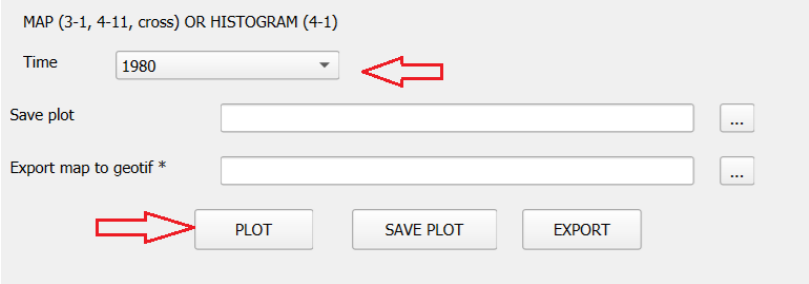


The maximum number of frost events in this area occurred in 1980. We will create a map for that year. It can also be observed that there is no clear trend in the number of frost days in May; however, a slight decrease is noticeable after 2011.

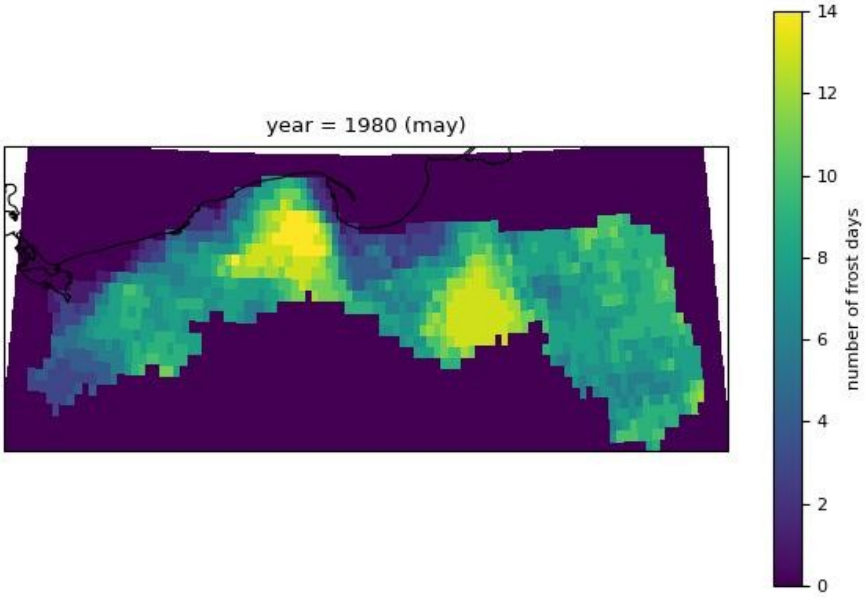
To create a map for 1980, we repeat the entire process, changing the expression (without the division).



In the middle part of the panel, we select the year 1980 and click “PLOT”.

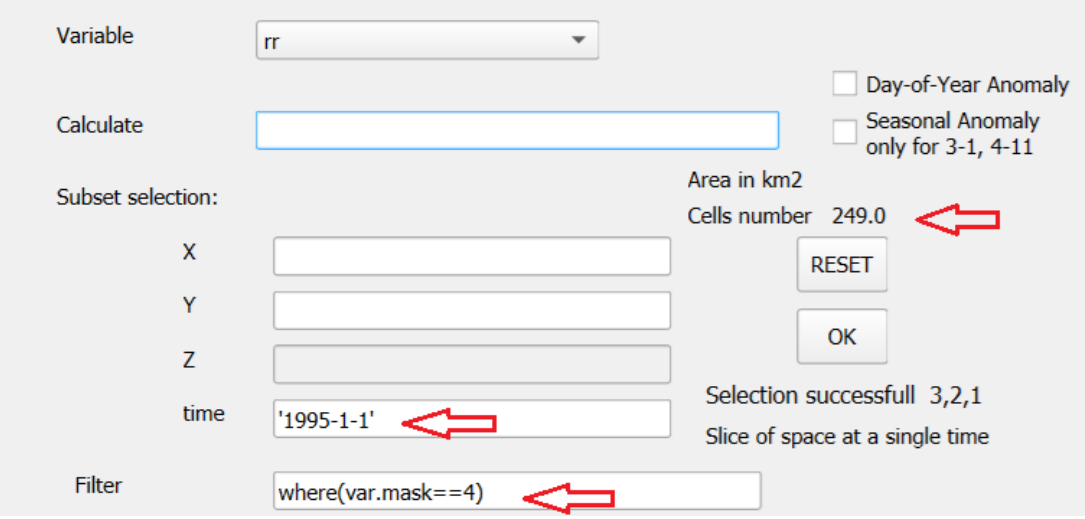


We obtain a map showing the distribution of the number of frost days in May 1980 (the axis labels have been changed in the figure properties).

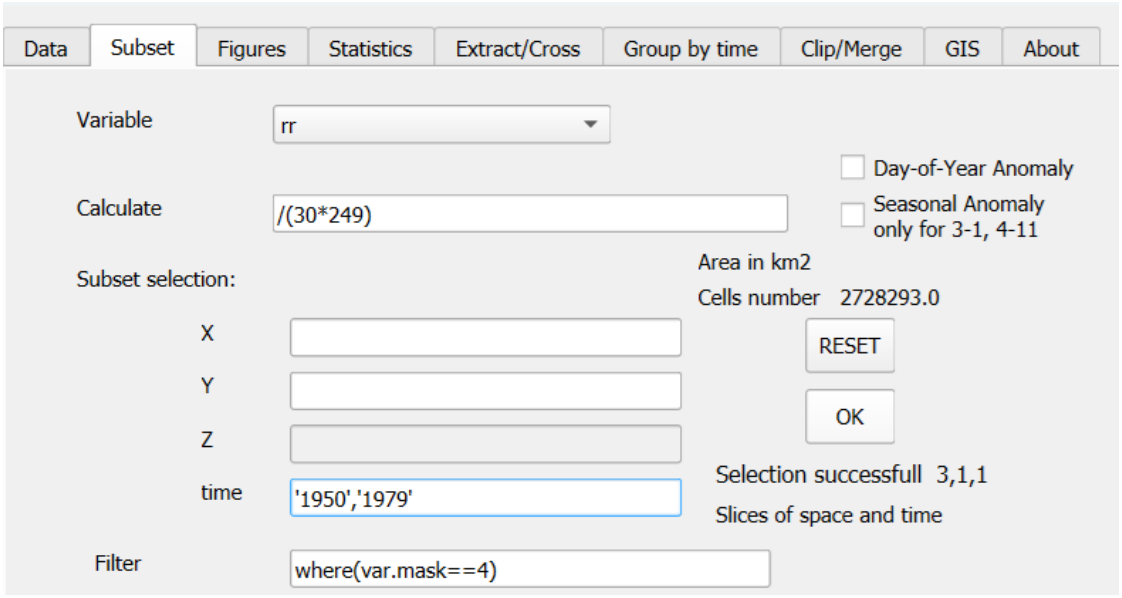


Changes in precipitation amounts by month
in the Pomeranian Voivodeship – comparison of periods 1950–1979
and 1995–2024

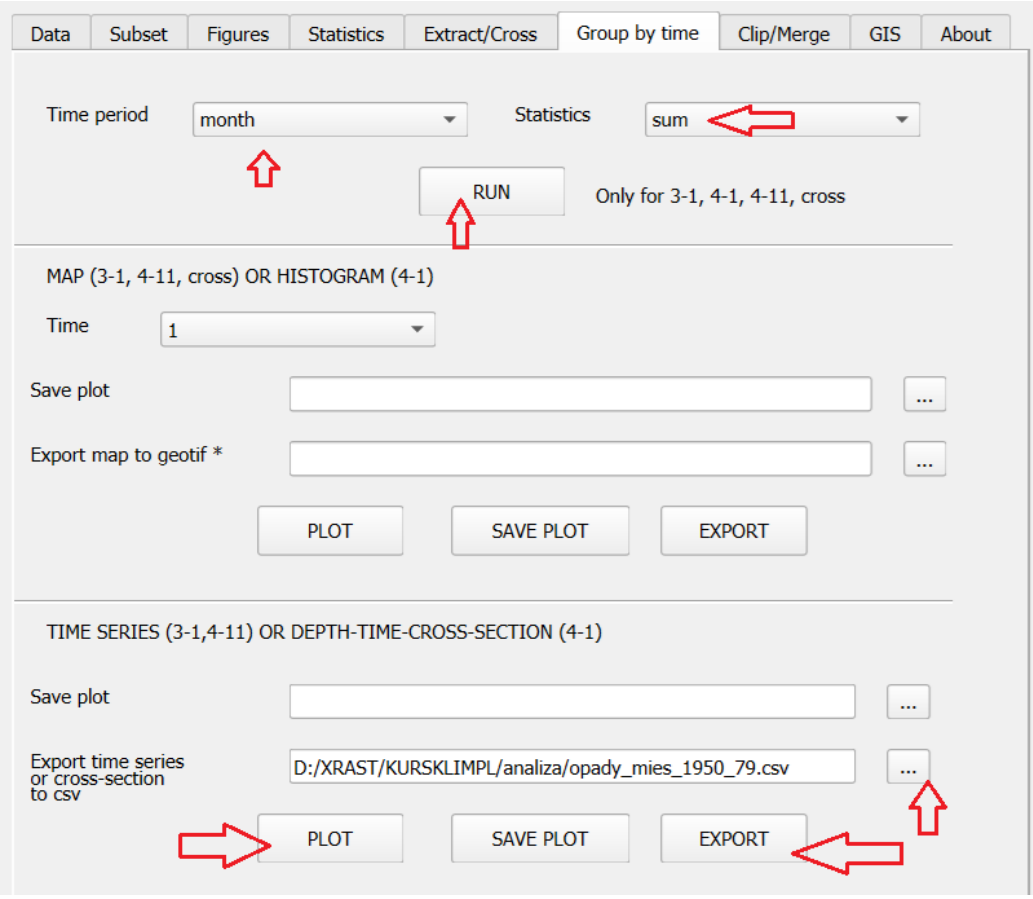
The Pomeranian Voivodeship has ID = 4. The precipitation data (rr) are expressed in [mm]. To calculate the average value for a given area over 30 months (a climatological period is 30 years), we divide the sum of values by the area of the voivodeship (which can be determined as in the previous query).



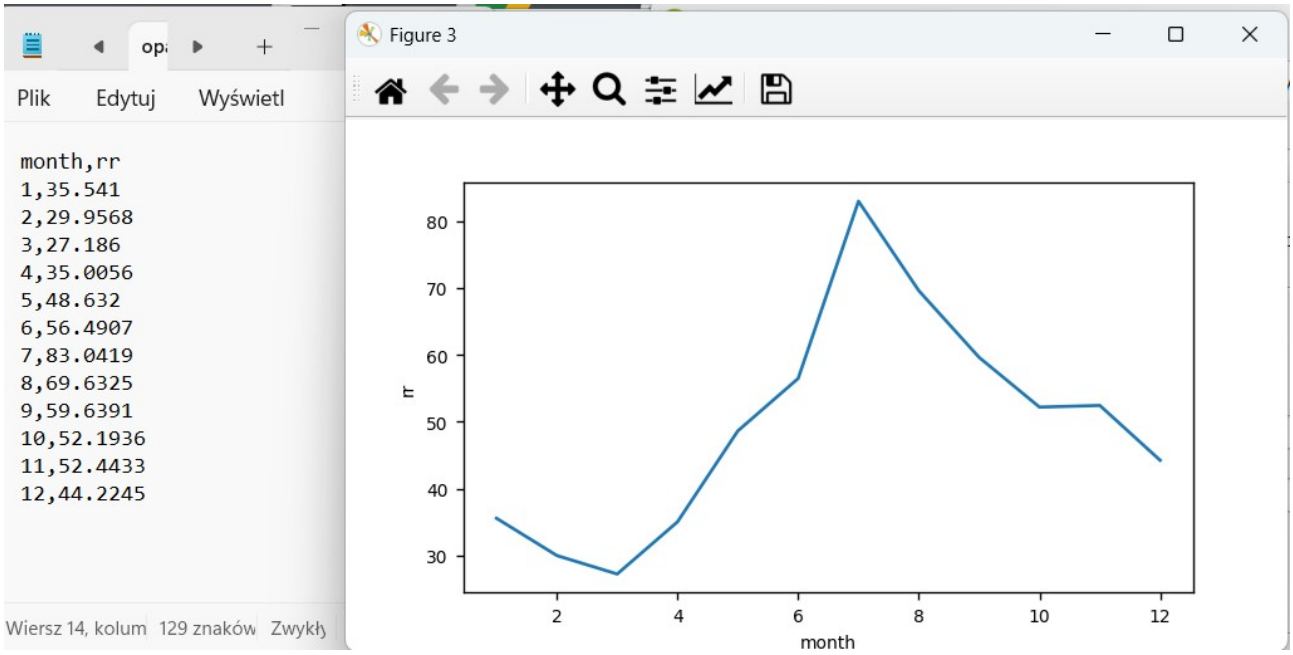
We divide the variable by (30 * 249), i.e., the number of grid cells summed for each month across the entire area. We create a subset using the expression:



Then we go to the **Group by time** tab, group the data by months (*Time period* = month) and sum all values (*Statistics* = sum). Since the values were divided by the number of cells in the monthly dataset, we obtain the average monthly precipitation in mm. Then we click RUN.

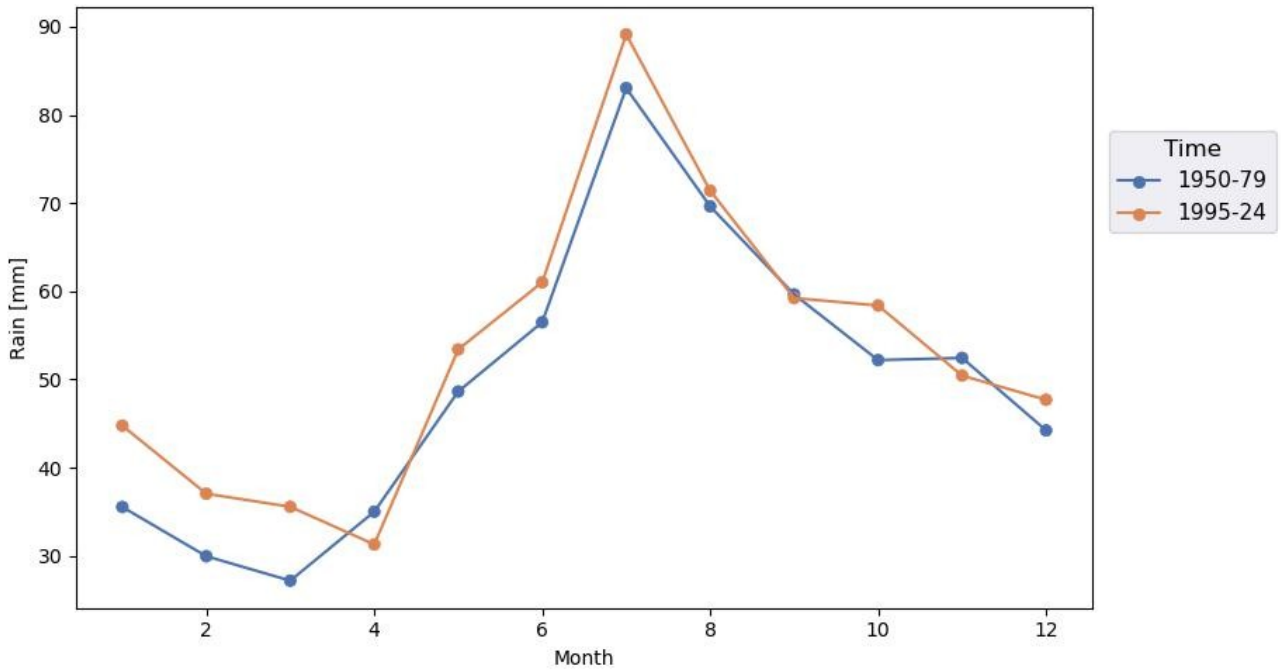


We go to the bottom of the panel, click PLOT, and obtain a graph showing precipitation variability by month. The data can be saved to a text file. We enter its name and click EXPORT.



We will perform the same sequence of operations for the 30-year period 1995–2024. As a result, we obtain two plots that are difficult to compare when not overlaid.

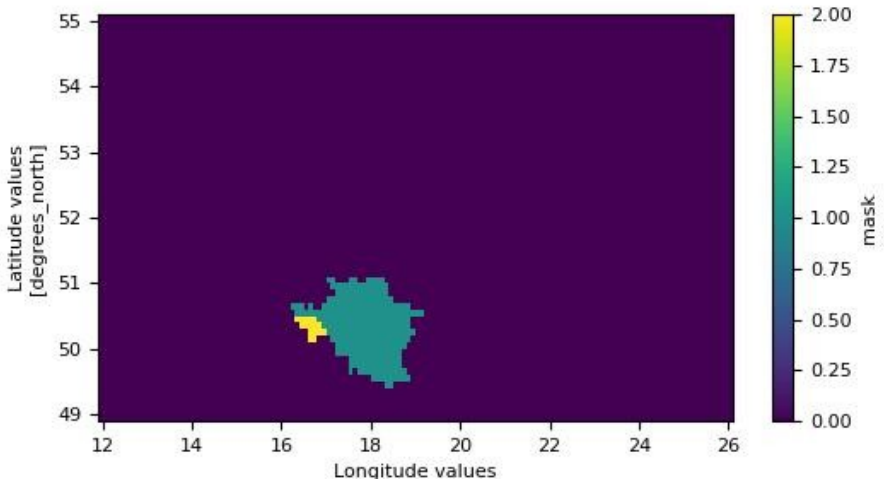
We will use the two saved text files with the results, which will allow us to create a combined plot outside the program.



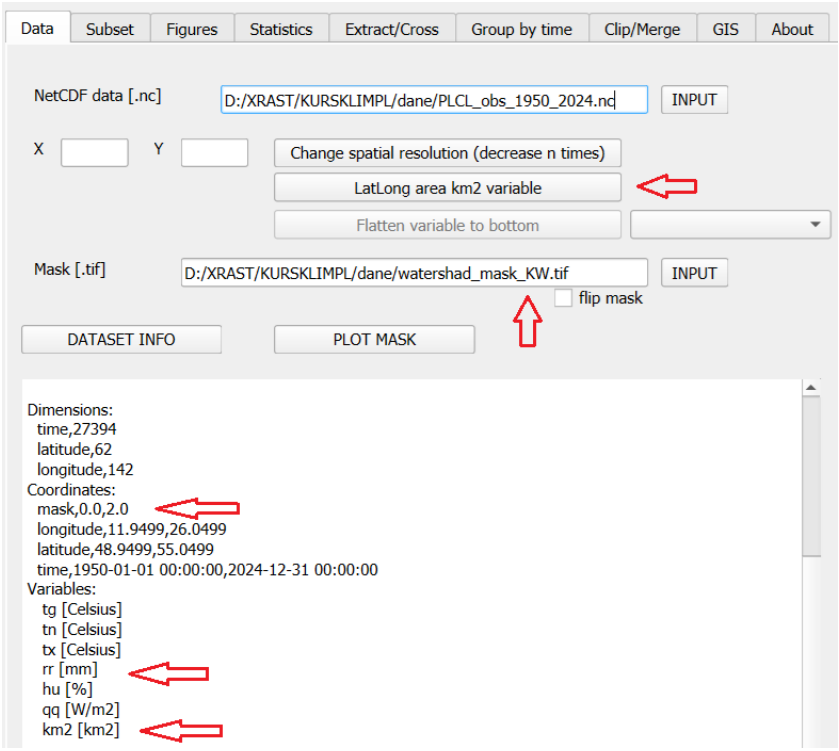
The figure was created using Python in a Jupyter Notebook. Examples of code for generating plots from text files exported from Xraster have been collected in four Jupyter notebooks (Xrast_plot_ipynb.zip).

Total precipitation over the Nysa Kłodzka catchment basin (in million m³) upstream of Kłodzko during the floods of 1997 and 2024

The first step will be to input the data, create a variable describing the raster cell area, and import the mask for the required catchment (it has ID = 2).

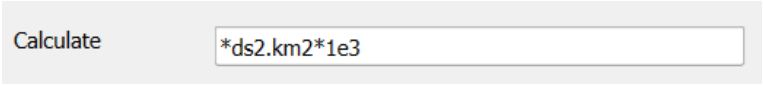


The data will be downloaded in the following format.

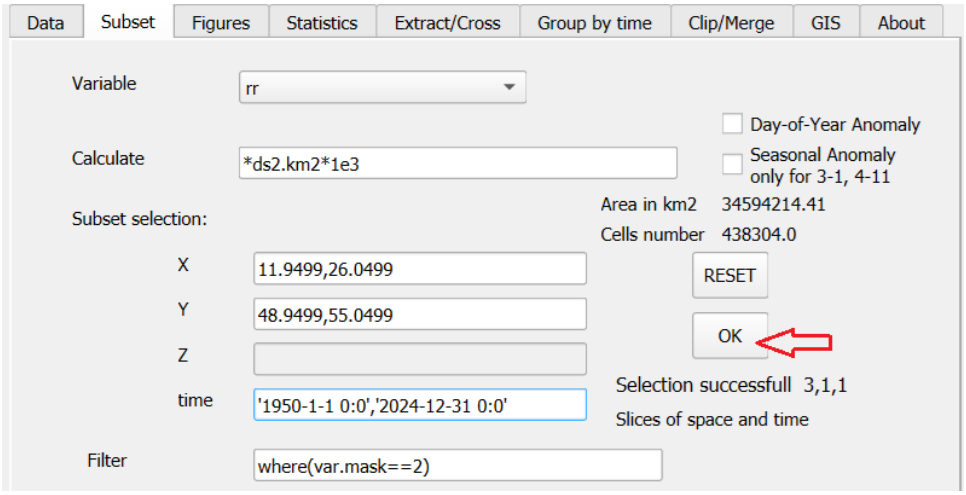


Click the **LatLong area km2 variable** button, which creates the **km2** variable containing the area of each cell in km².

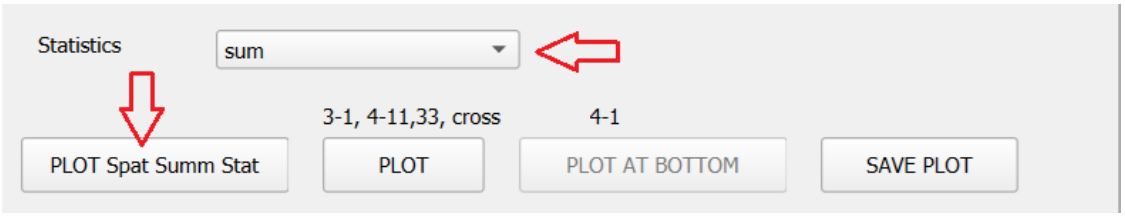
The **rr** variable represents the daily total precipitation per raster cell in **[mm]**. If the raster cell area is described by the **km2** variable in **[km²]**, then the expression calculating the volume of water that fell on a catchment cell on a given day, using the selected **rr** variable, is as follows:



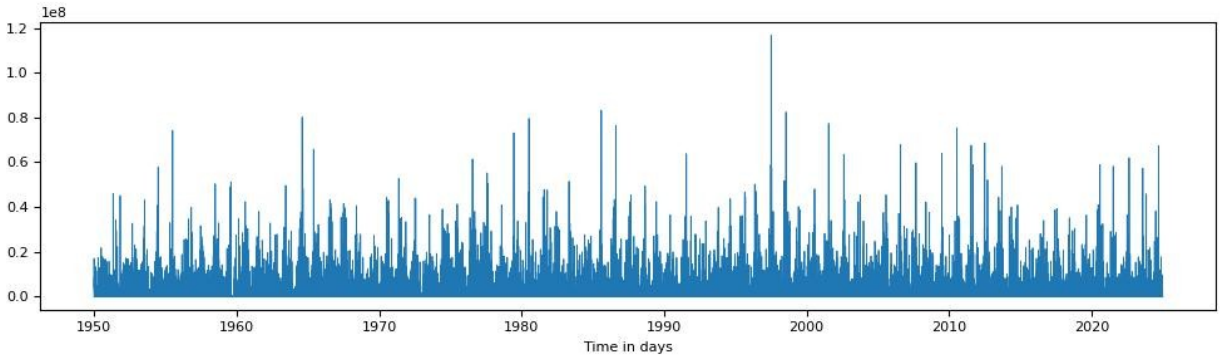
We create a subset using this expression for the catchment area (**mask == 2**).



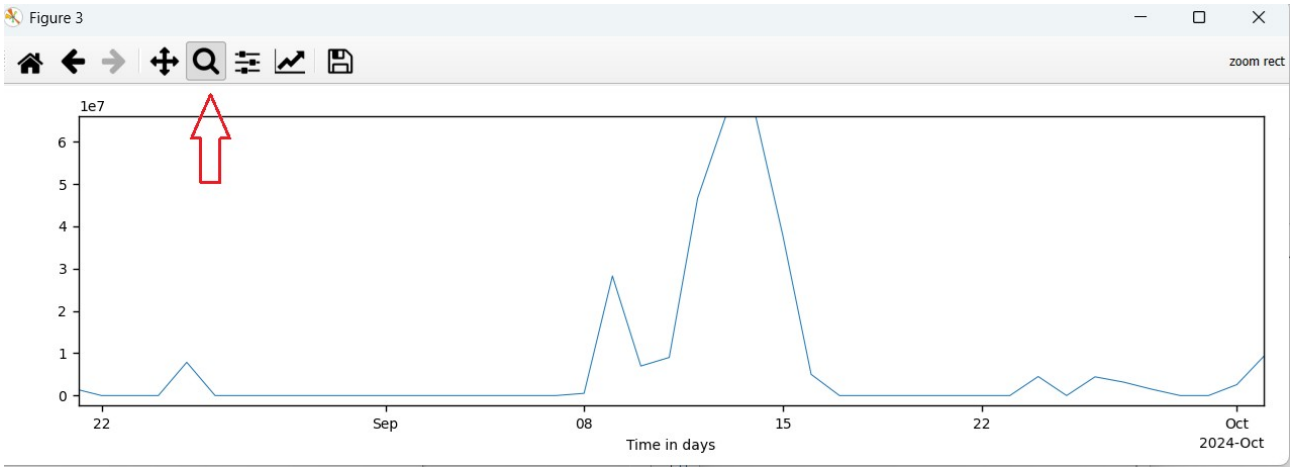
We go to the **Statistics** tab.



Click the **PLOT Spat Summ Stat** button with the **Statistics** setting set to **sum**. This will produce a time series plot of the total precipitation over the catchment for each day.

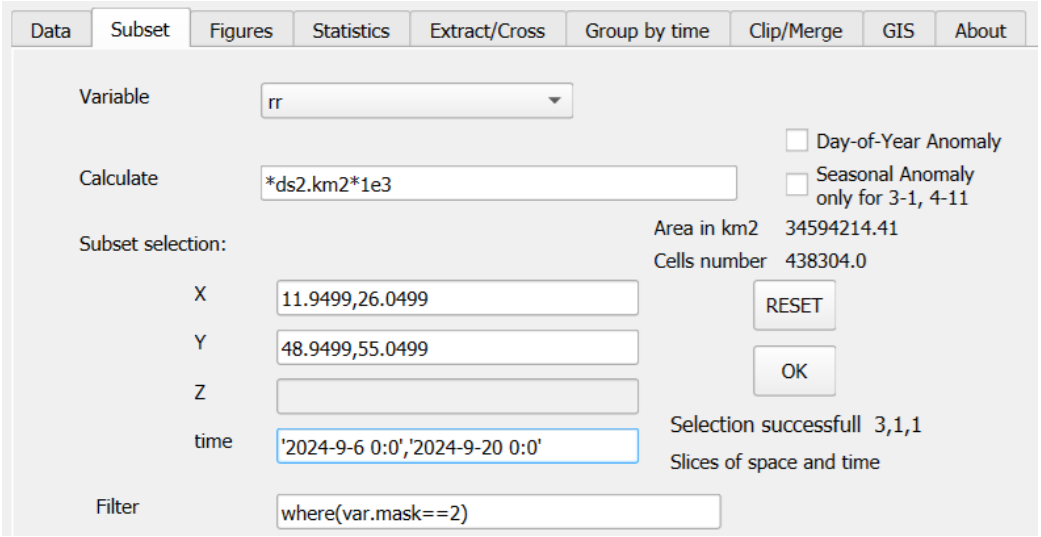


The plot in the figure window can be enlarged to analyze the time periods of interest.

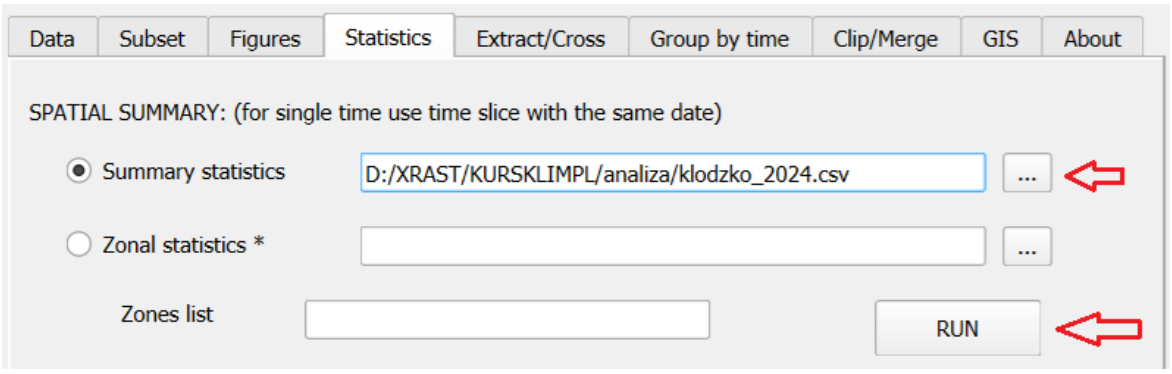


We define the time periods for flood analysis in 1997 and 2024:
1997-06-29 to 1997-07-13
2024-09-06 to 2024-09-20

We create a subset for each period. For example, for 2024.



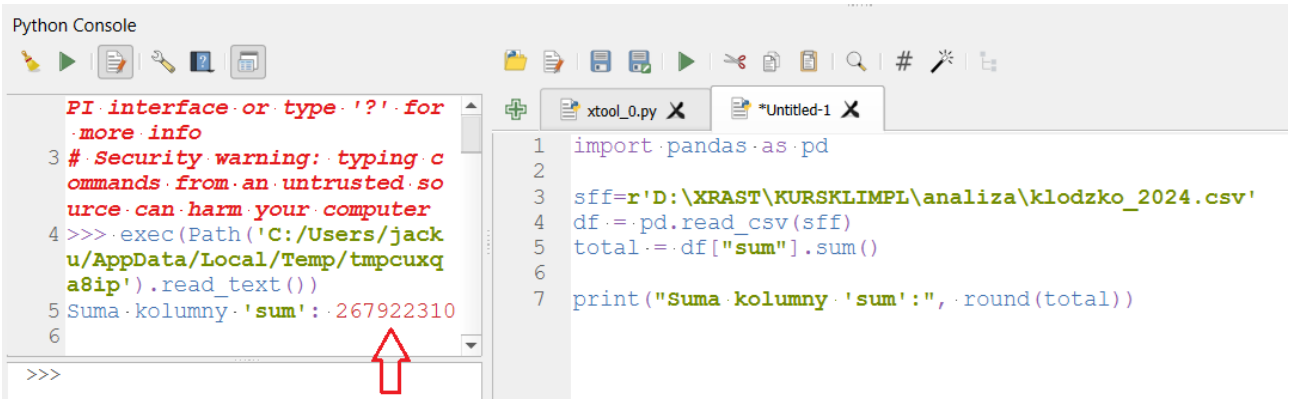
We go to the **Statistics** tab and run the **Summary statistics** tool, which creates a text file containing statistics for all cells in the subset (the catchment defined by the mask) as well as the total sum of those values (the total precipitation over the catchment in m^3).



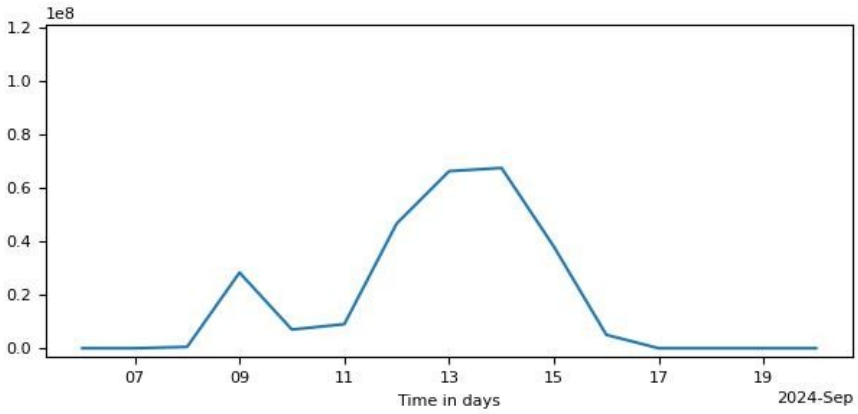
The file has the following format,

```
time,mean,min,max,median,std,sum
2024-09-06 00:00:00,0.0,0.0,0.0,0.0,0.0, 0.0
2024-09-07 00:00:00,0.0,0.0,0.0,0.0,0.0, 0.0
2024-09-08 00:00:00,33002.5418,0.0,267682.4935,0.0,87326.1173, 528040.67
2024-09-09 00:00:00,1767350.7568,1254453.0872,2527237.7687,1731467.7486,307101.6084, 28277612.11
2024-09-10 00:00:00,436521.9197,229761.9205,623281.7077,433456.4332,108230.0569, 6984350.72
2024-09-11 00:00:00,563118.2137,284624.8393,1062856.9297,492063.4122,214262.8643, 9009891.42
2024-09-12 00:00:00,2914248.855,2440634.4312,4055903.738,2837286.5564,385361.1365, 46627981.68
2024-09-13 00:00:00,4136190.9791,2558729.6456,8088089.1022,3510417.4262,1413326.4695, 66179055.67
2024-09-14 00:00:00,4210487.2735,2579912.9203,9471682.0158,3617097.7625,1756553.2123, 67367796.38
2024-09-15 00:00:00,2370550.744,1901477.9001,2714036.8907,2355012.9741,254764.7493, 37928811.9
2024-09-16 00:00:00,313673.1063,110919.5423,497047.4468,315950.9562,98573.3457, 5018769.7
2024-09-17 00:00:00,0.0,0.0,0.0,0.0,0.0, 0.0
2024-09-18 00:00:00,0.0,0.0,0.0,0.0,0.0, 0.0
2024-09-19 00:00:00,0.0,0.0,0.0,0.0,0.0, 0.0
2024-09-20 00:00:00,0.0,0.0,0.0,0.0,0.0, 0.0
```

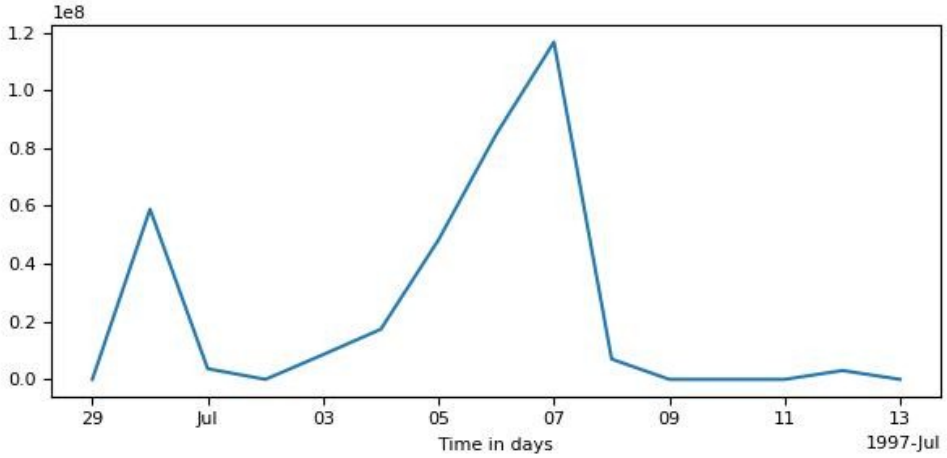
In the Python console in QGIS, you can sum the **sum** column using a simple script.



The total precipitation volume over the catchment during the 2024 flood was **267.9 million m³**. The time series of this flood obtained using the **PLOT Spat Summ Stat** button will have the following form,



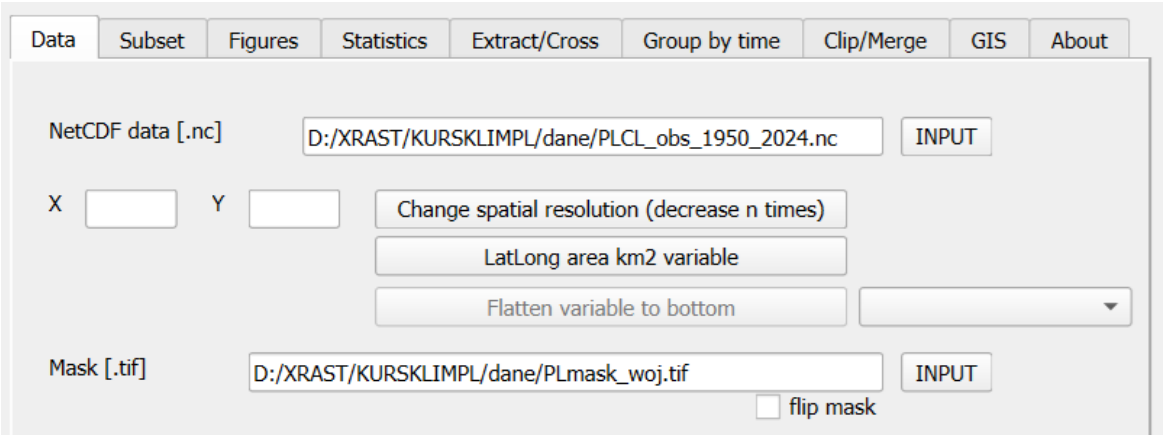
By repeating these steps for the year 1997, we obtain a total precipitation of **348.3 million m³**, as well as a plot.



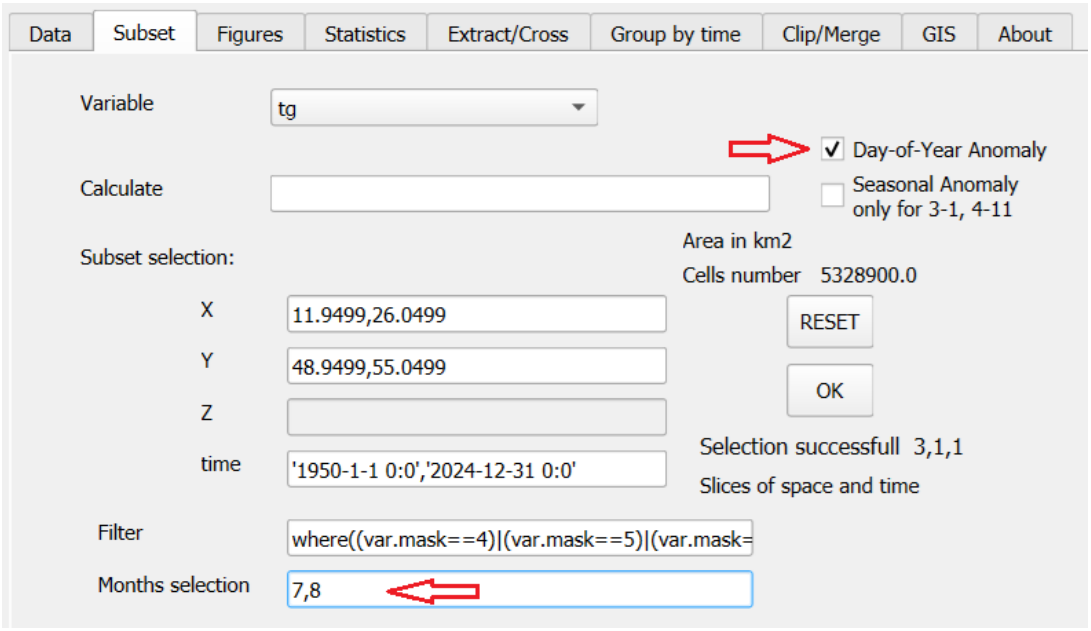
Daily temperature anomaly maps of northern Polish voivodeships – years with the highest positive and negative anomalies .

Anomalies will be calculated as deviations from the mean daily temperature for each day of the year at each grid cell. Maps of anomalies for the days with the largest positive and negative anomalies will be presented.

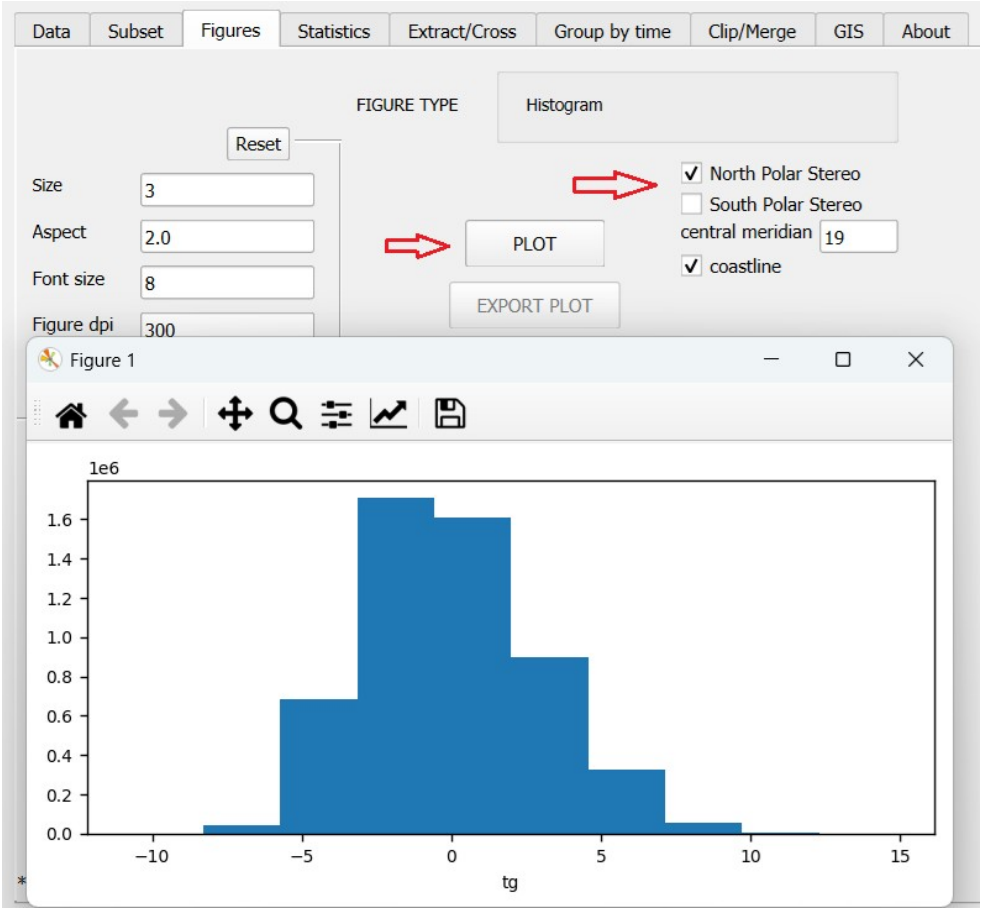
We load the data and the mask.



We create a subset of the daily mean temperature anomaly variable (selecting **Day-of-Year Anomaly**) and restrict the subset to July and August.

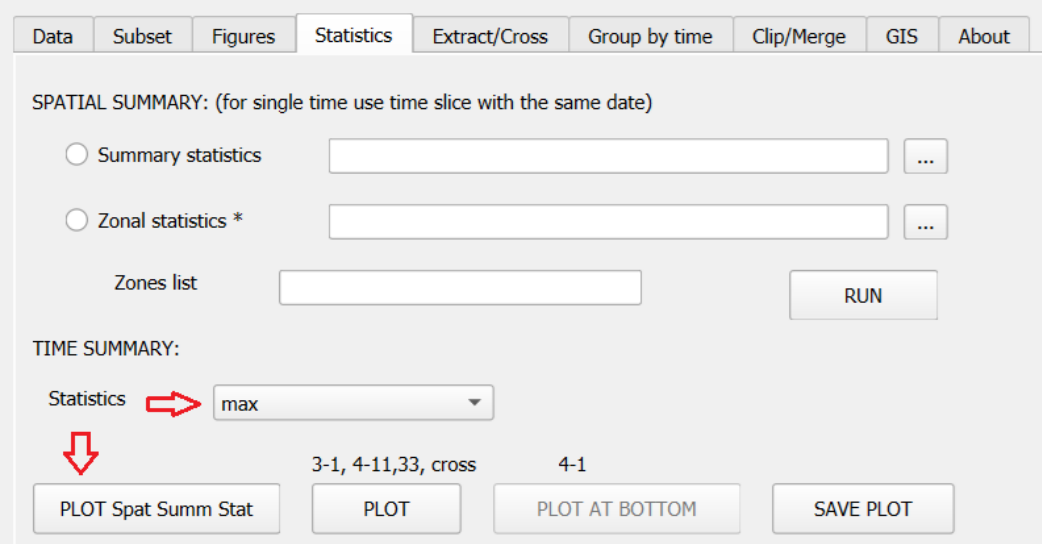


In the **Figures** tab, we create a histogram of the anomalies. At the same time, we add a coastline and use the **North Polar Stereographic** projection with a central meridian of **19°**.

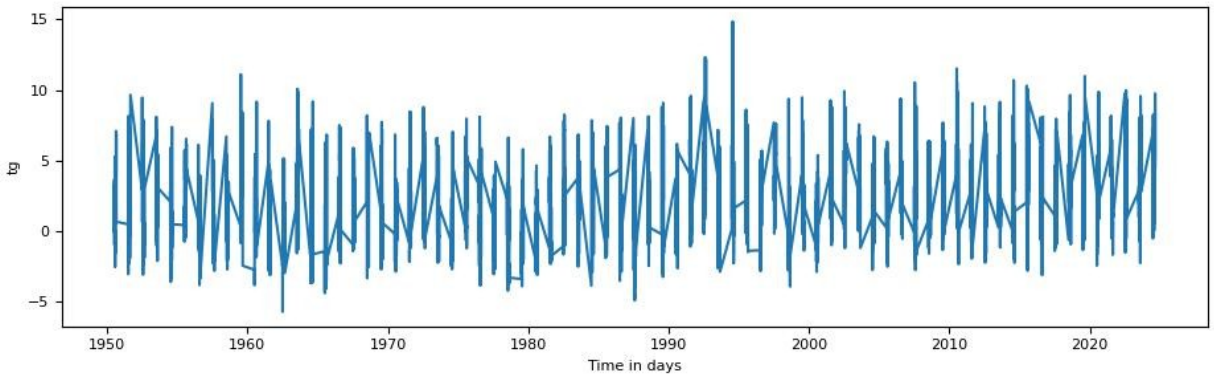


We can see that the maximum positive and negative anomalies exceed 10 degrees (parts of the histogram should be zoomed in using the magnifying tool).

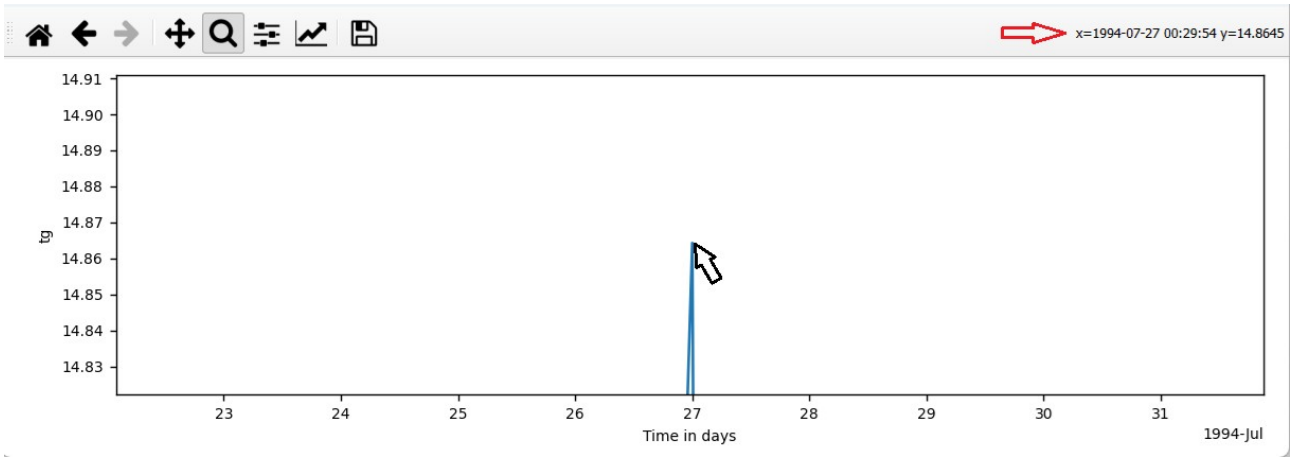
Next, we go to the **Statistics** tab and create time series of the maximum and minimum anomaly values. First, we select the **Max** statistic and click **PLOT Spat Summ Stat**. Then, we repeat the procedure for the **Min** statistic.



For the **Max** statistic, the time series has the following form:

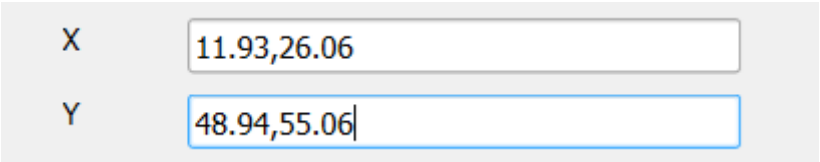


We can zoom in on a portion of the figure to read the date of the maximum value in 1994.

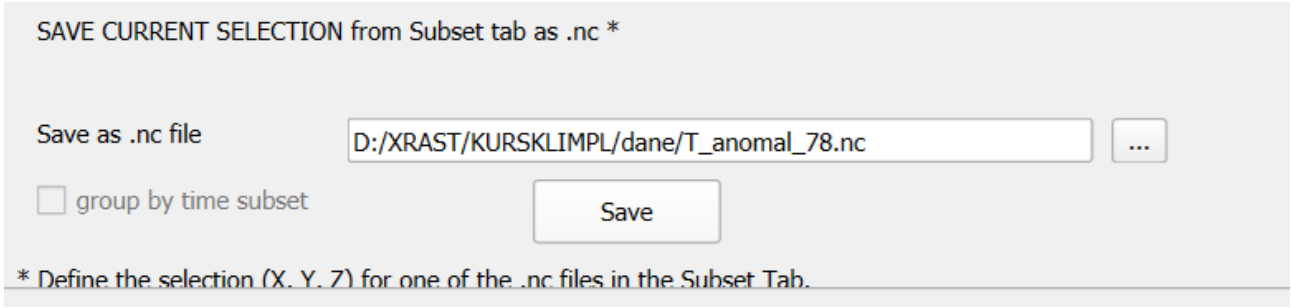


The maximum positive anomaly occurred on 27-07-1994 (14.86). In a similar way, we can find the largest negative anomaly (using the **Min** statistic). It occurred on 15-07-1993.

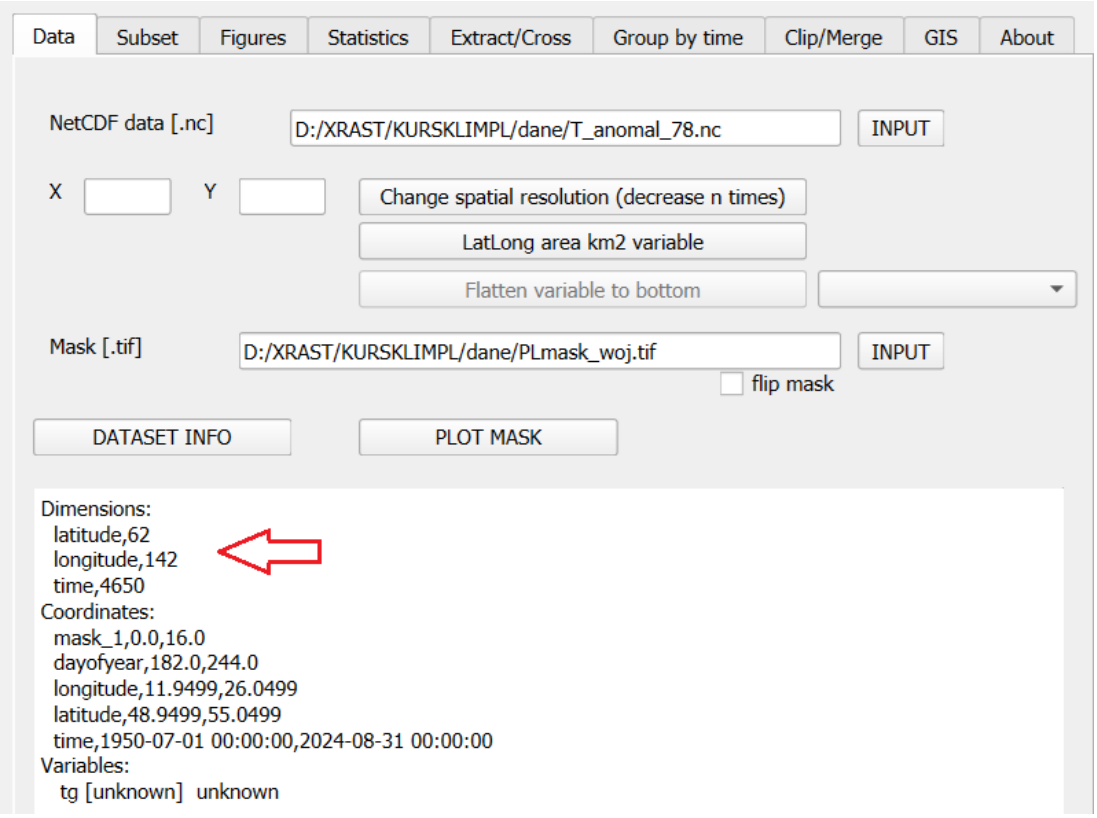
To access individual anomaly maps, we need to save them to a new subset. We change the x and y value ranges (in the second position after the dot) to obtain the same raster size.



Next, we go to the **Clip/Merge** tab and save the current subset.

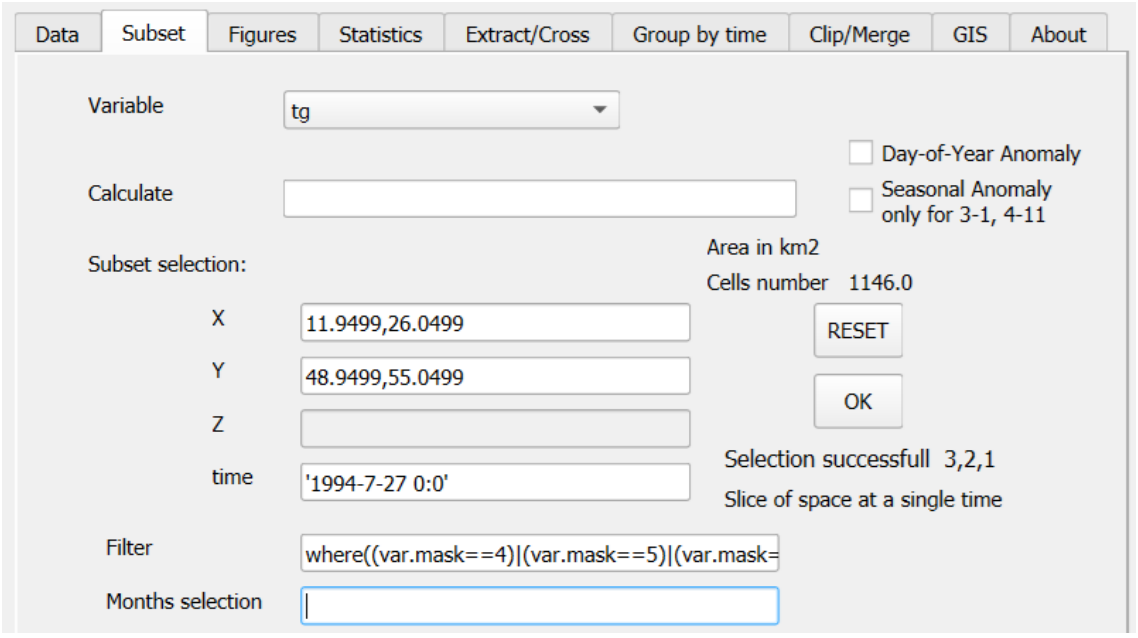


We load this dataset and the map of voivodeships. We check whether the raster size (number of rows and columns) is identical. If it were not, it would not be possible to apply the mask.

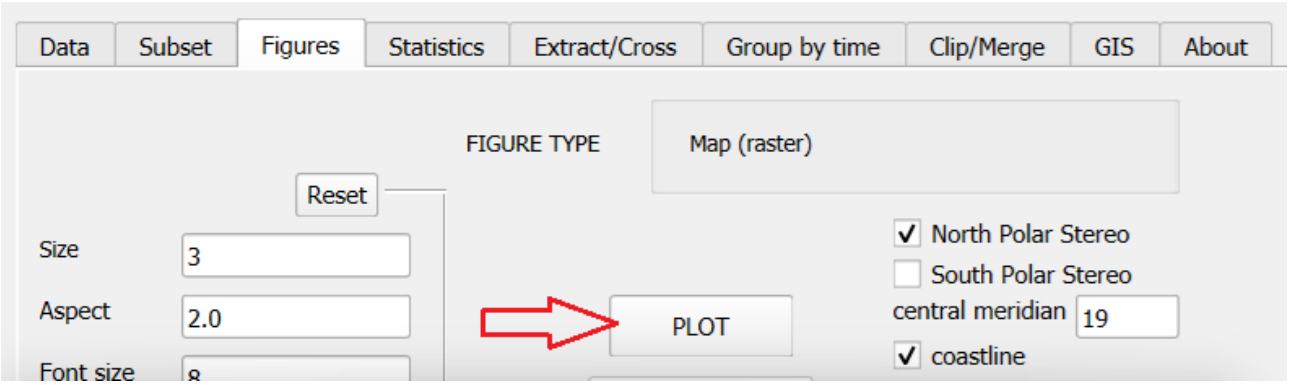


In this dataset containing anomalies of daily mean temperature, we have access to individual daily cases.

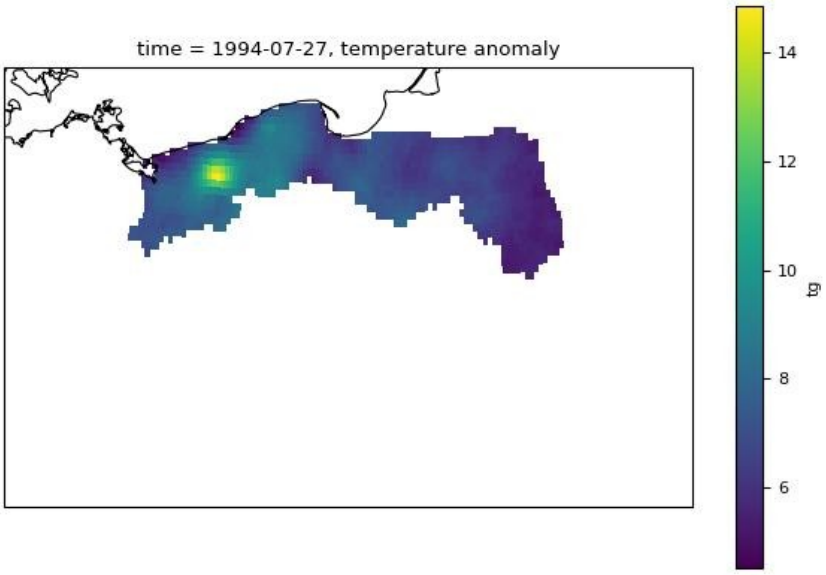
We create a subset of anomalies for 27-07-1994.



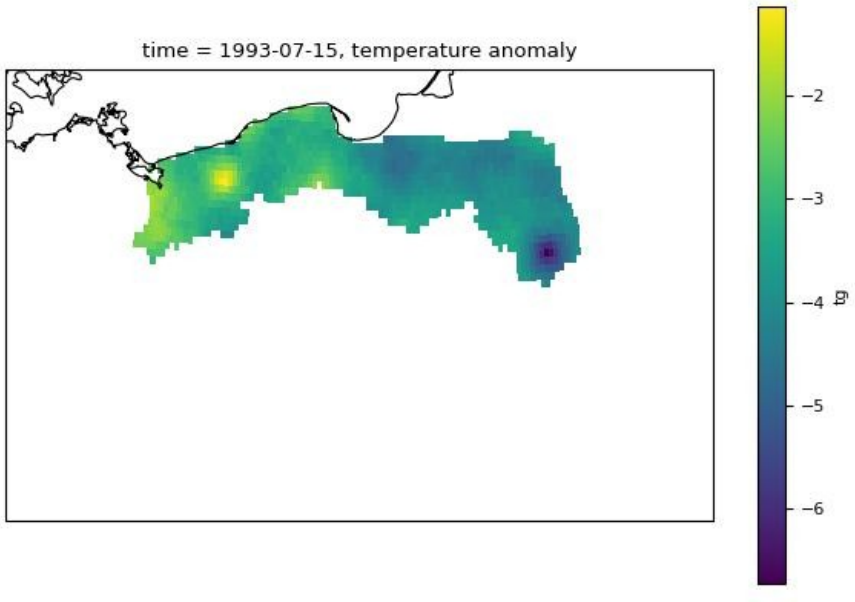
Next, we go to the **Figures** tab.



The created map is shown below.



We perform a similar analysis for 15-07-1993 to identify the situation in which the largest negative anomaly occurred. As a result, we obtain a map.

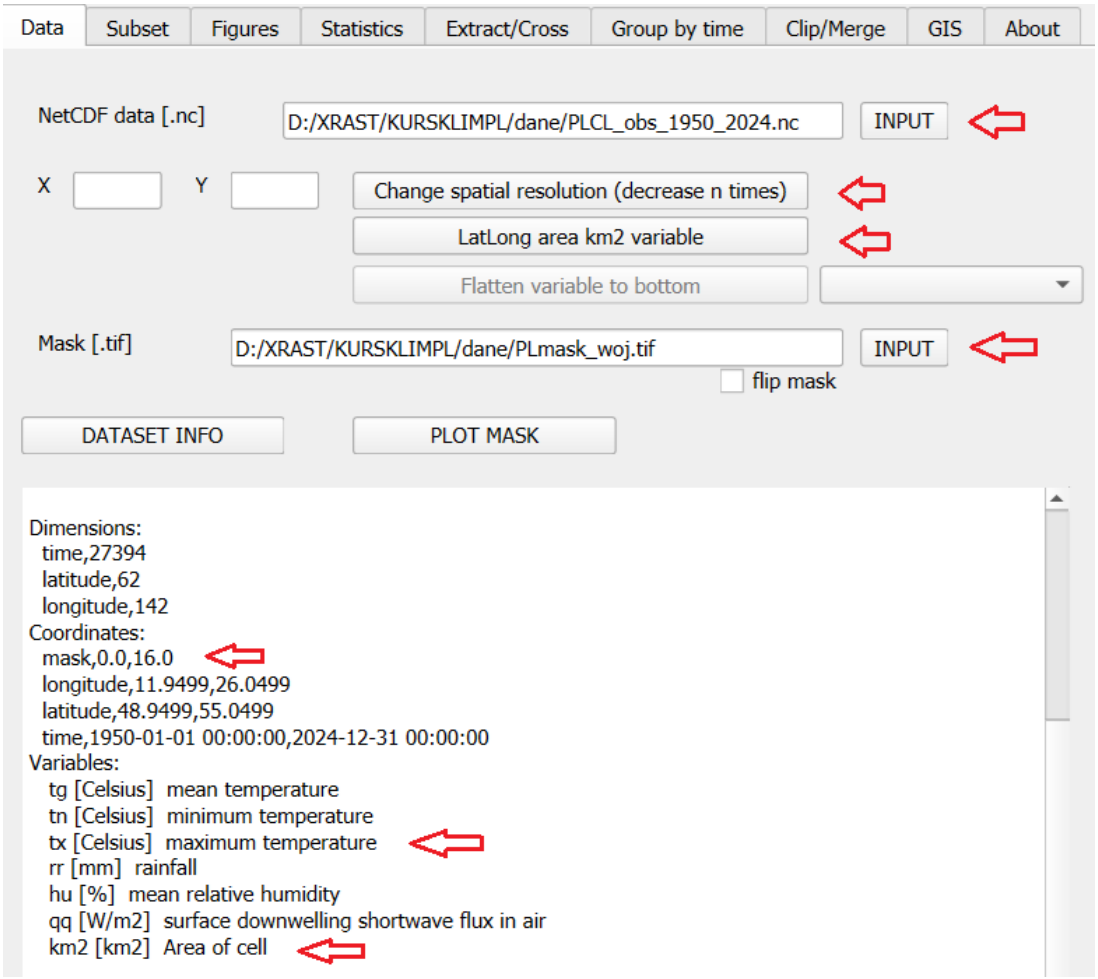


Analysis of the largest heatwaves in Poland over the last 75 years

There are many methods for defining heat waves. In this task, we will use the methodology from the Polish scientific literature, defined as follows: A heat wave (HW) is defined as a sequence of at least three consecutive hot days, with a maximum temperature of at least 30°C, occurring at at least 10 out of 46 meteorological stations in Poland. This means that heat affected more than 20% of the country.

Our goal will be to describe the three largest heat waves that occurred in Poland over the last 75 years, taking into account both their spatial extent and duration.

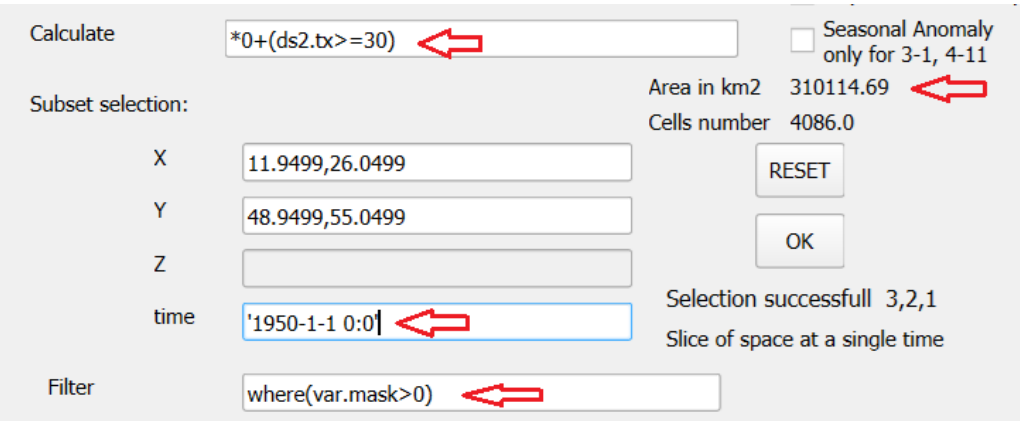
We begin by importing the data and creating variables for grid-cell area (in km²) and the voivodeship mask.



We create a subset of the maximum temperature transformed using the expression:

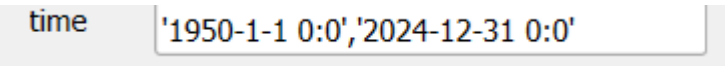
```
0 + (ds2.tx >= 30)
```

First, we will determine the threshold area (20% of the country's area). We will perform this calculation for any single day, applying the mask to define the area of Poland.

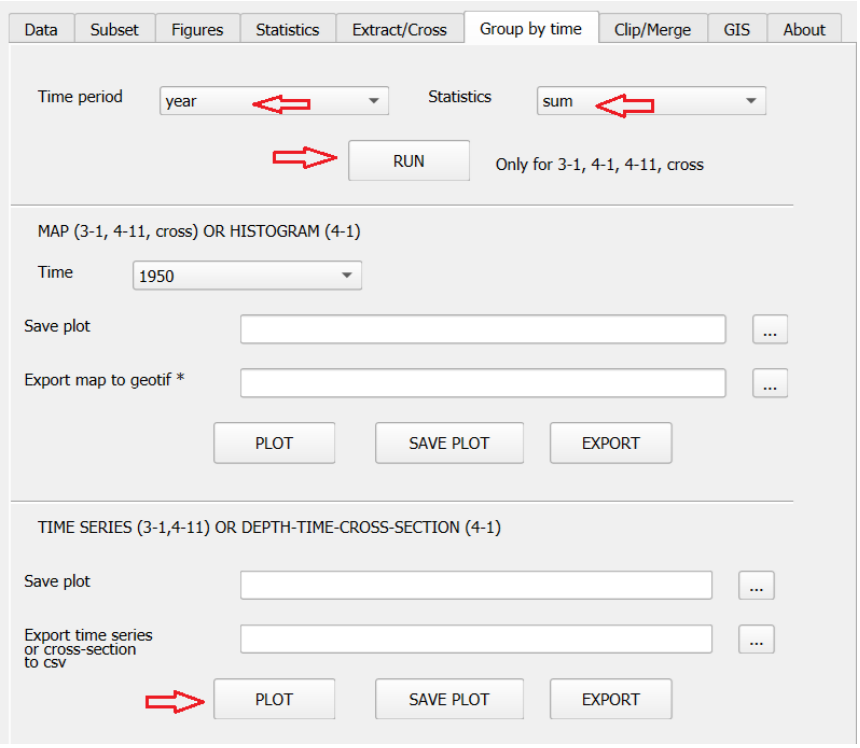


The modeled area of Poland consists of 4,086 grid cells (310,000 km²), so 20% of the country corresponds to approximately 80 grid cells (62,000 km²).

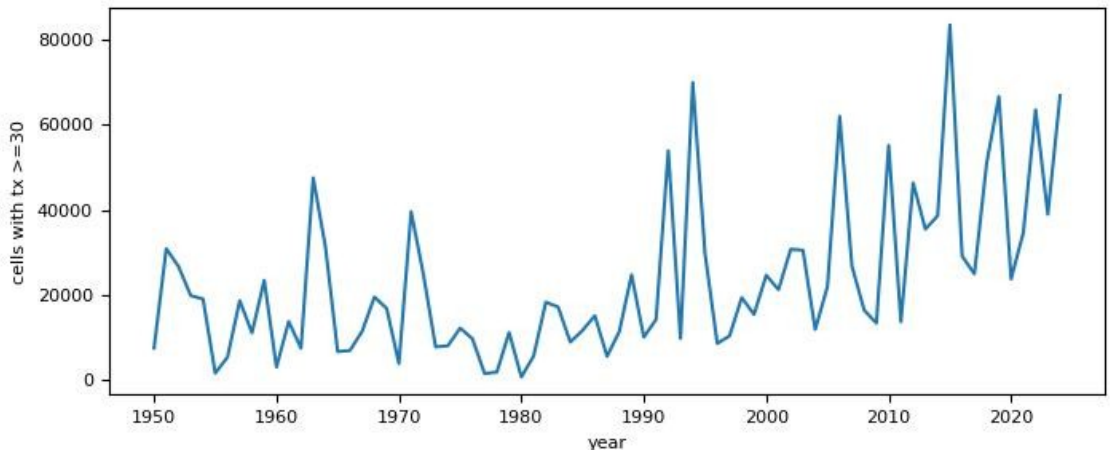
We repeat this operation for the entire time period.



We can then sum the grid cells meeting the above condition separately for each year to obtain a time-series plot showing the total number of grid cells with a maximum temperature of at least 30°C.

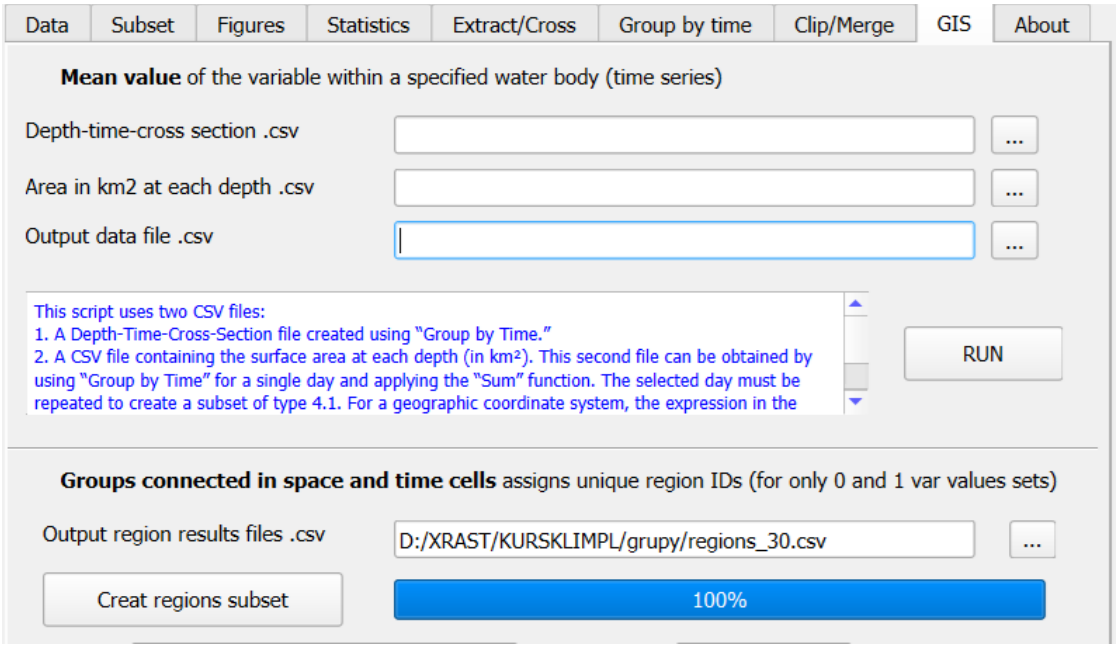


The resulting plot will have the following form:



After 1990, there is a visible increasing trend in the frequency of temperatures ≥30°C, expressed as the sum of grid cells exceeding this threshold.

Heat waves are defined as groups of cells connected in space and time with a value of 1. We then go to the **GIS** panel, enter the output file name for the clustering result, and click the **Create regions subset** button.



As a result, a text file named ***name+sum*** (regions_30_sum.csv) is created, which—after sorting by the area of the regions—has the following form:

id,max_number_of_cells,date_start,date_end,date_of_max

21668,4059,1992-08-06,1992-08-12,1992-08-10

42228,4058,2024-07-09,2024-07-17,2024-07-10

22797,4052,1994-07-21,1994-08-09,1994-07-31

38583,4041,2019-06-24,2019-06-27,2019-06-26

35844,4037,2015-08-02,2015-08-17,2015-08-07

30628,4033,2007-07-14,2007-07-22,2007-07-16

10898,3997,1971-07-24,1971-08-08,1971-08-07

6805,3986,1963-08-01,1963-08-08,1963-08-04

6752,3982,1963-07-23,1963-07-26,1963-07-24

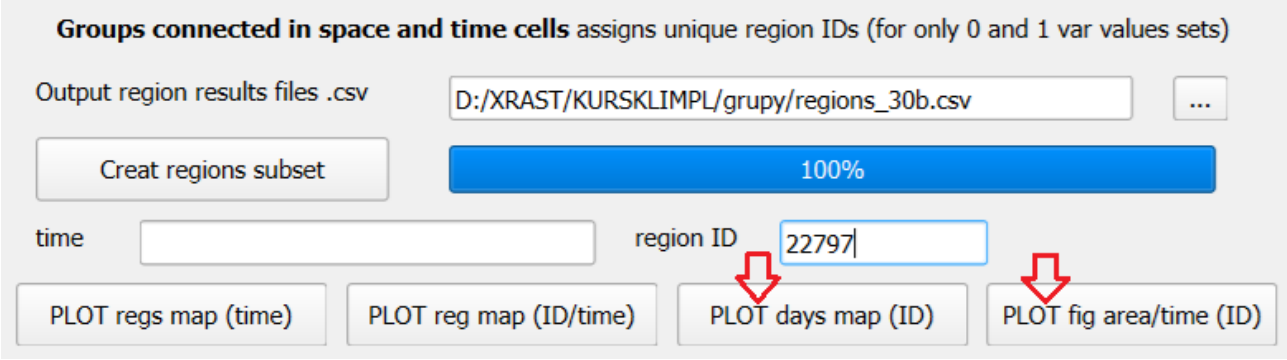
36378,3971,2016-06-23,2016-06-27,2016-06-25

25059,3957,1998-07-20,1998-07-24,1998-07-21

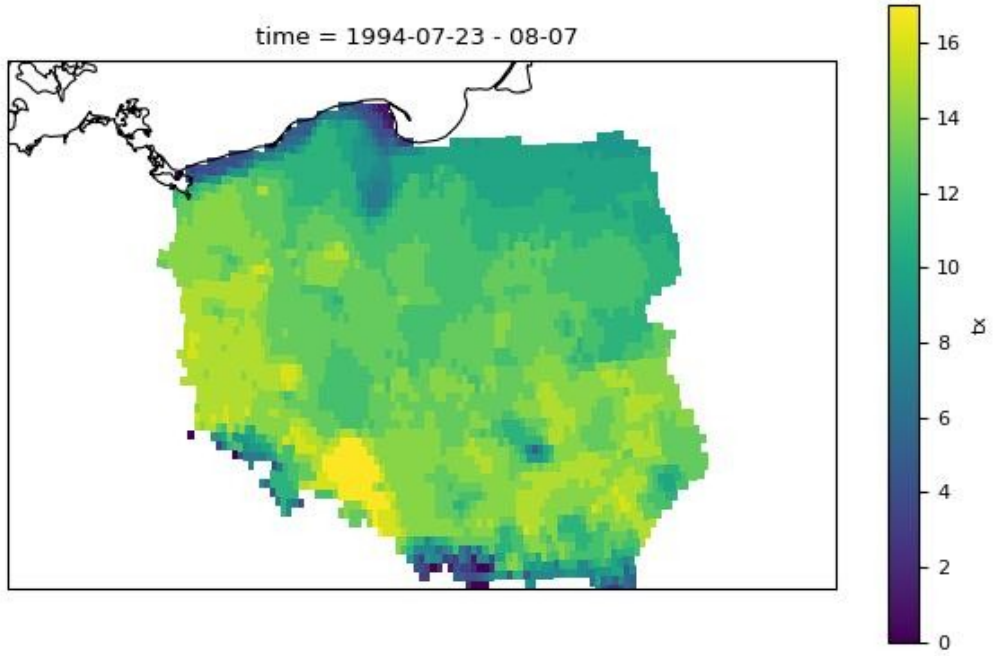
We will consider the duration of heat waves, since the maximum spatial extents are similar and close to the threshold representing the total area of Poland (4,086 grid cells). We will analyse the heat waves from 21-07-1994 to 09-08-1994 (20 days), 02-08-2015 to 17-08-2015 (15 days), and 24-07-1971 to 07-08-1971 (15 days). The condition of exceeding 62,000 km² will be verified during the analysis.

The first column of the file contains region identifiers (spatio-temporal groups). We insert it into the **region ID** field and can obtain two outputs: a map of the number of days in a given region and a plot of the cumulative area of the region over time, which allows us to verify the actual number of days (with area above 62,000 km²).

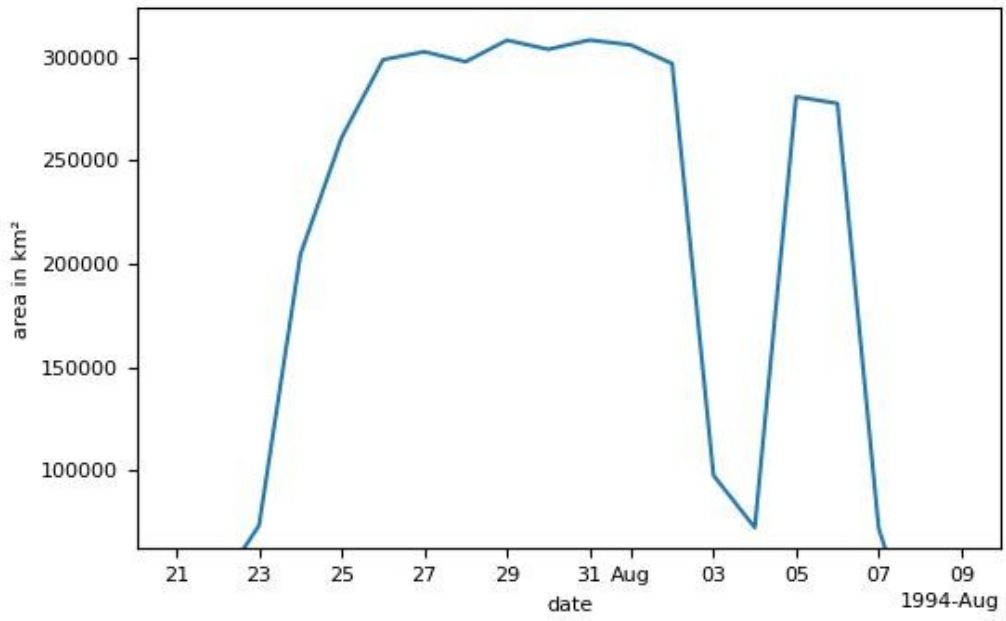
On the plots, the vertical axis was adjusted so that it starts from this threshold value.



The map of the number of days will have the form,



The map allows for a precise determination of how the heat wave was distributed across Poland. It can be exported as a GeoTIFF raster for further processing in QGIS. The time series plot of the area allows for correcting the duration of the heat wave.



It can be read that the duration was 16 days. The area values from the time series can be saved to a text file.

4. Creating a Multidimensional Dataset for Sardinia

–

As an example of creating a dataset for a specific area, we will use data engineering tools to create a dataset containing **daily mean temperature and precipitation for Sardinia for the period 1995–2024**.

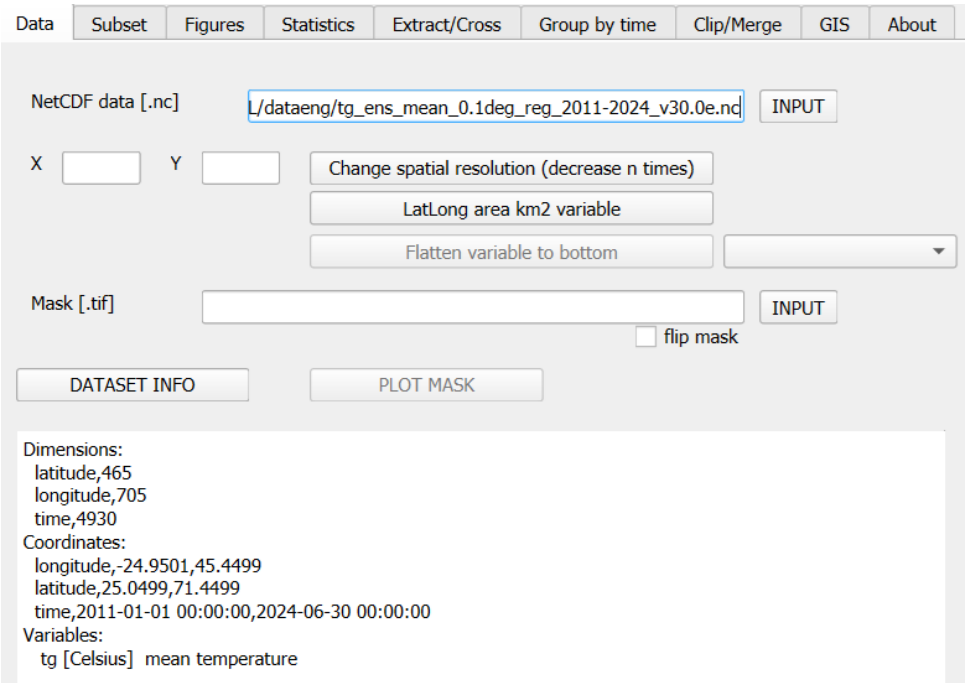
From the Copernicus Data Store, we download data from the dataset
***E-OBS daily gridded meteorological data for Europe from 1950 to present derived from in-situ observations*.**

The following datasets were downloaded. **Separate datasets are available for different time periods and variables.**

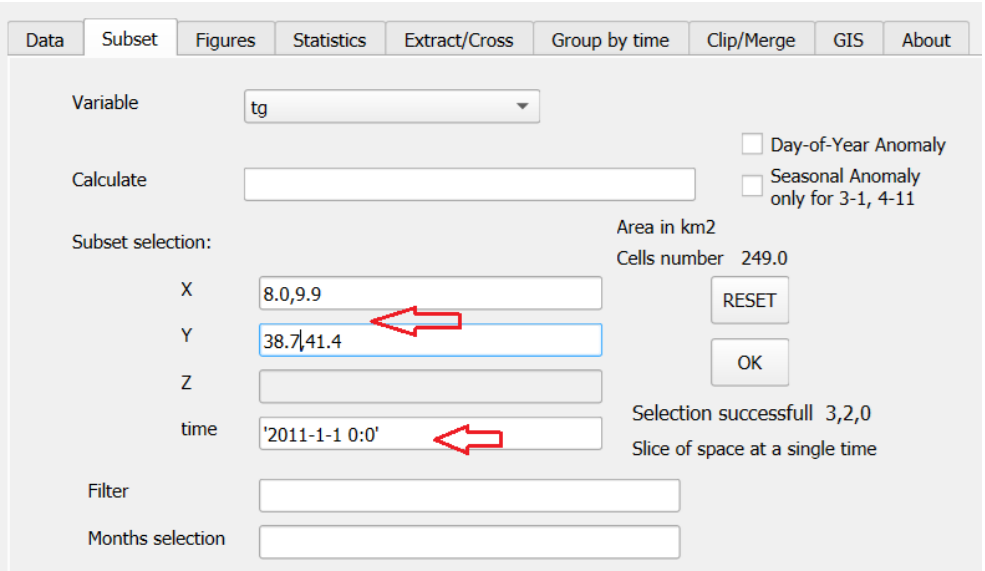


Here, **rr** represents precipitation, while **tg** represents mean temperature. The first step is to determine the spatial extent of the data for Sardinia.

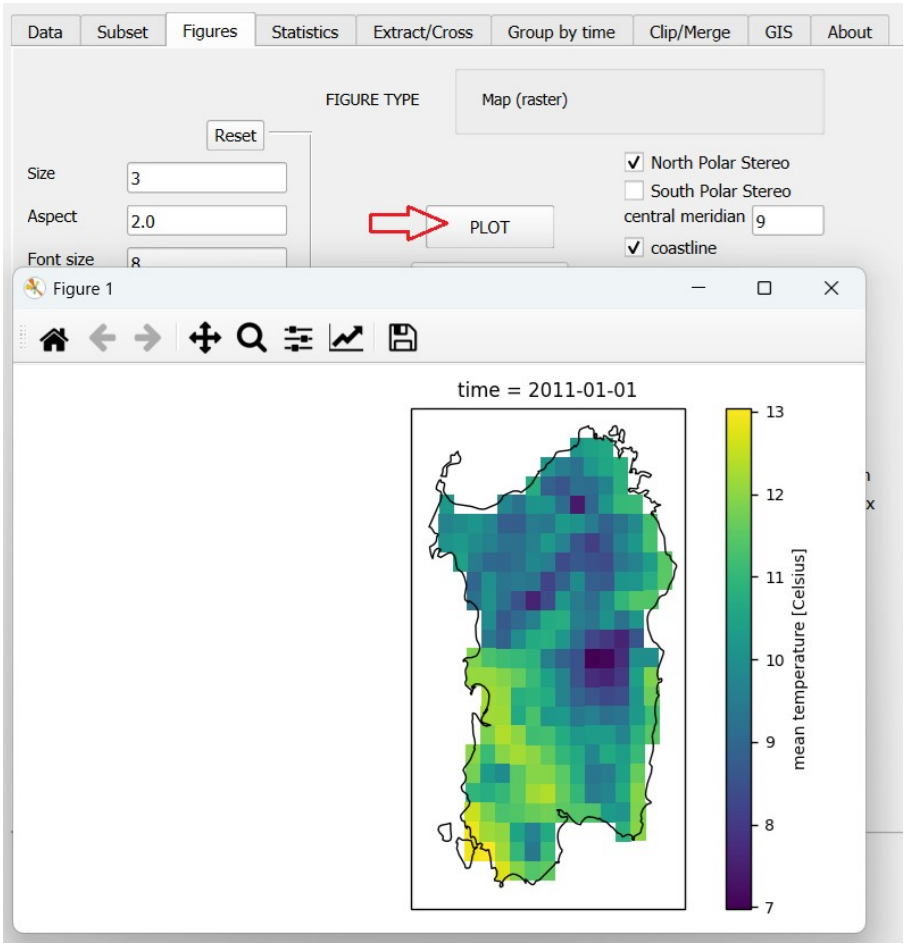
We select any one of the downloaded datasets.



Next, we enter the new coordinate range and create a subset for any selected day.



We verify the selection by creating a temperature map.



If the coordinate range is correct, we can proceed to the **Data Engineering** tools in the **Clip/Merge** tab.

We prepare two text files containing the full paths to the datasets for the same variable.

Dla tg : tg_sarda.csv

D:\XRAST\KURSKLIMPL\dataeng\tg_ens_mean_0.1deg_reg_1995-2010_v30.0e.nc

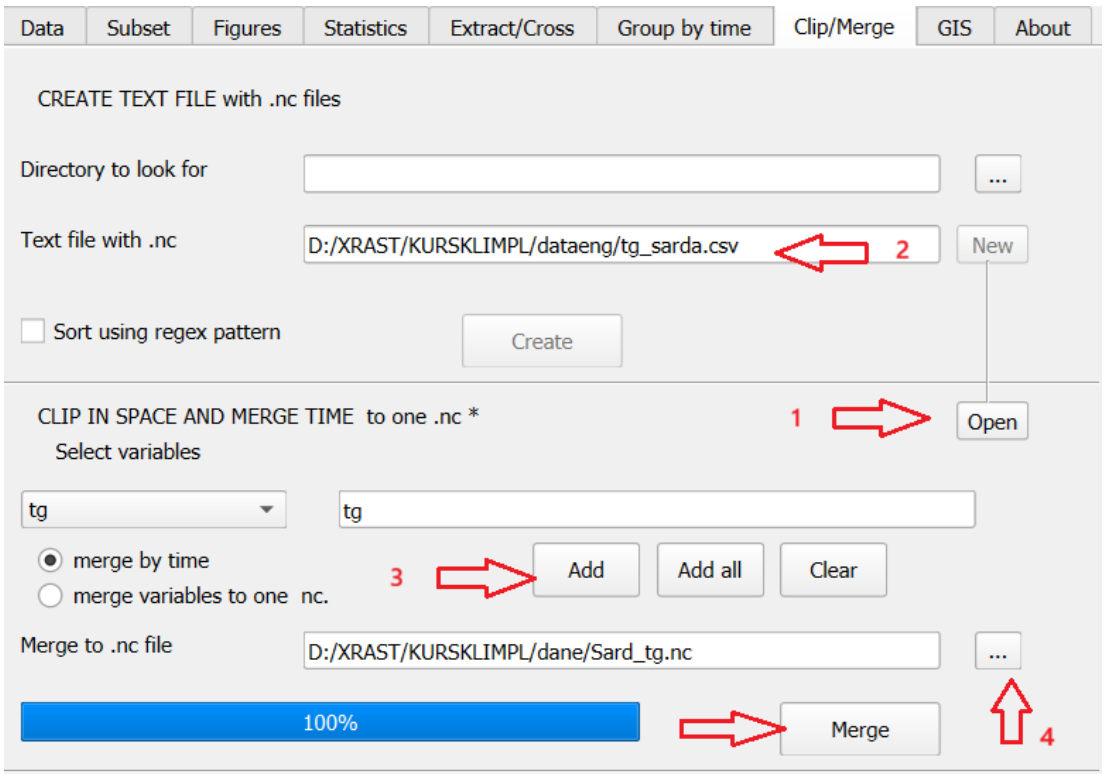
D:\XRAST\KURSKLIMPL\dataeng\tg_ens_mean_0.1deg_reg_2011-2024_v30.0e.nc

a dla rr: rr_sarda.csv

D:\XRAST\KURSKLIMPL\dataeng\rr_ens_mean_0.1deg_reg_1995-2010_v30.0e.nc

D:\XRAST\KURSKLIMPL\dataeng\rr_ens_mean_0.1deg_reg_2011-2024_v30.0e.nc

Przechodzimy do karty Clip/Marge i kolejno wpisujemy,



Open the text file **tg_sarda.csv**.

1. The files with the **.nc** extension appear in the **Text file with .nc** window.
2. Add the variable **tg** (there is only one variable).
3. Enter the name of the output file.

Click **Merge**. The progress bar must reach **100%**.

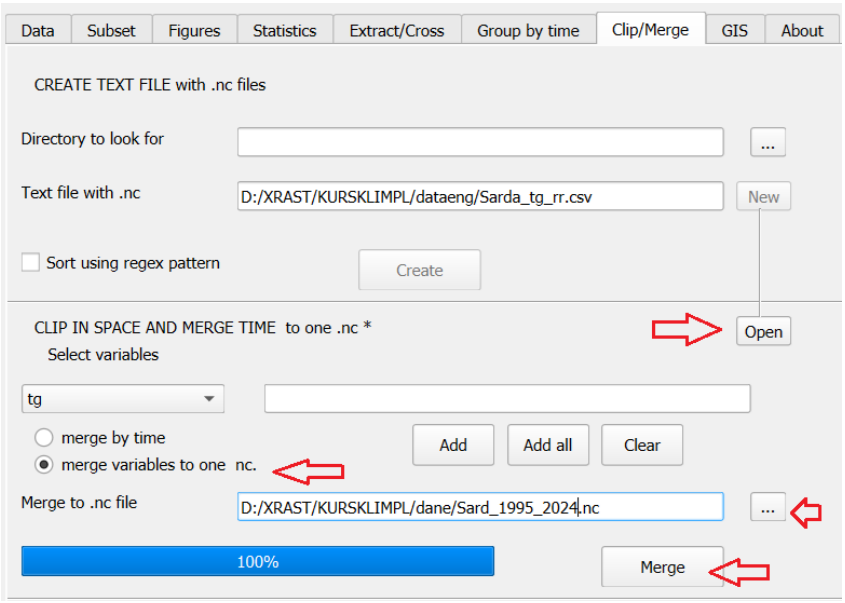
Repeat the same procedure for the **rr_sarda.csv** text file, creating **Sard_rr.nc**.

Next, we create a new text file containing the file paths for both variables, **tg** and **rr**, named **Sarda_tg_rr.csv**.

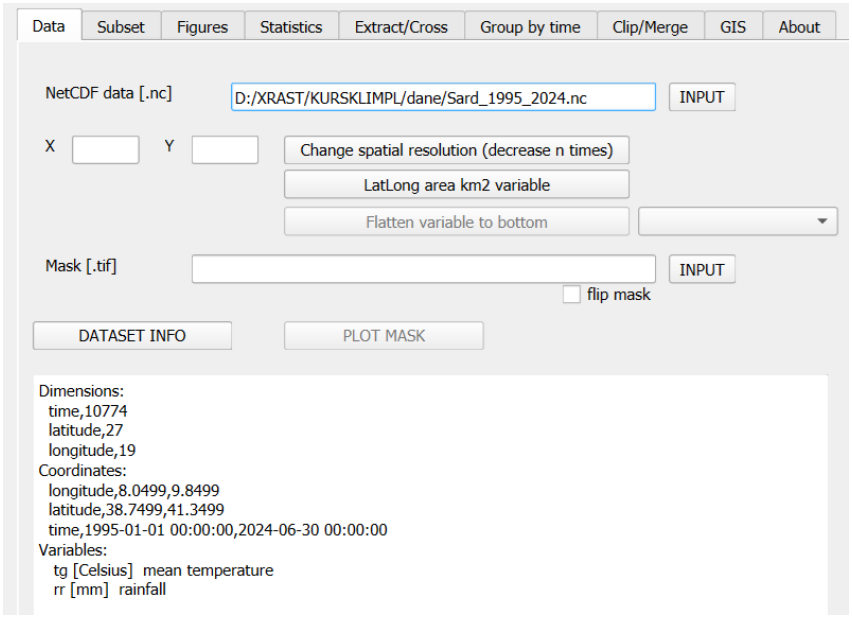
D:\XRAST\KURSKLIMPL\dane\Sard_tg.nc

D:\XRAST\KURSKLIMPL\dane\Sard_rr.nc

Next, we enter the file name, select the **Merge variables to one .nc** option, enter the name of the output file (**Sard_1995_2024.nc**), and click **Merge**.



We obtain a dataset covering only Sardinia and containing two variables: **daily mean temperature and precipitation**.

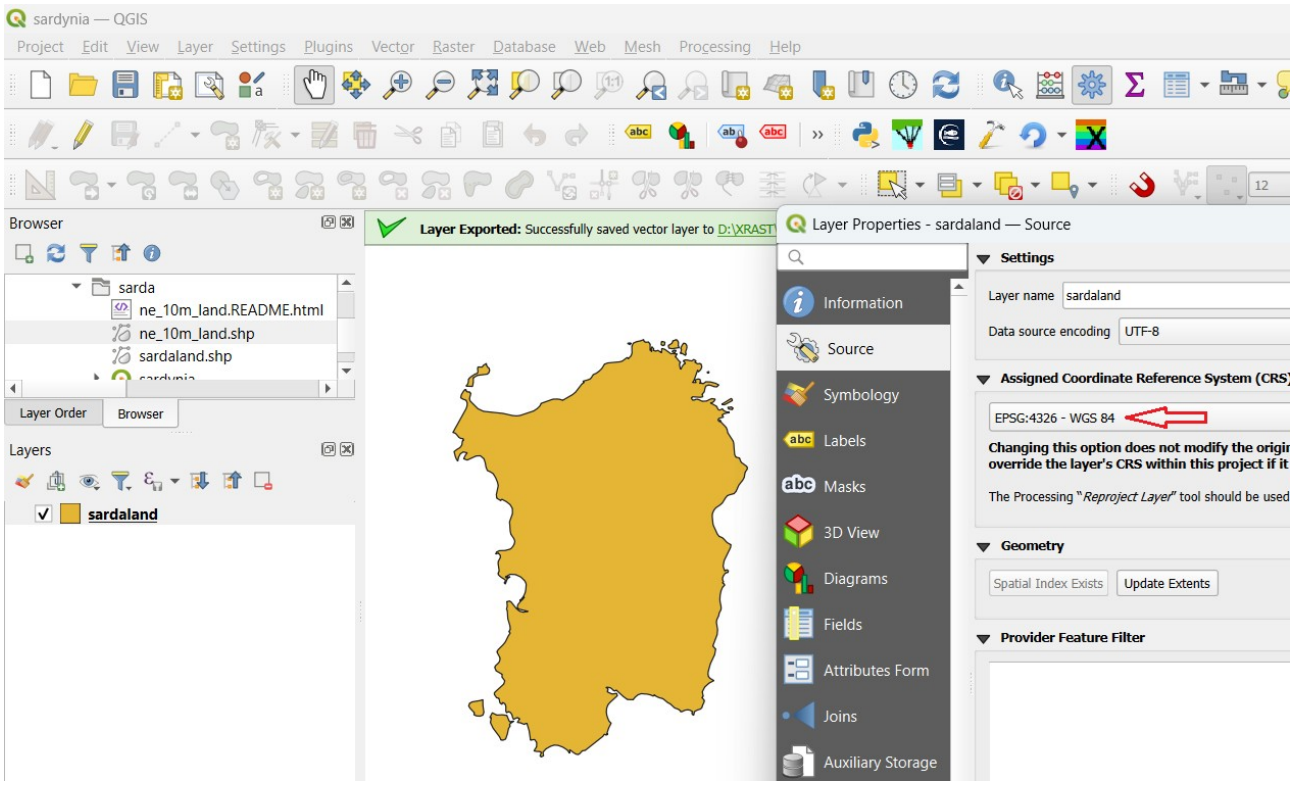


It should be noted that the data for **2024 are available only through the end of June**.

5. Mask for Sardinia

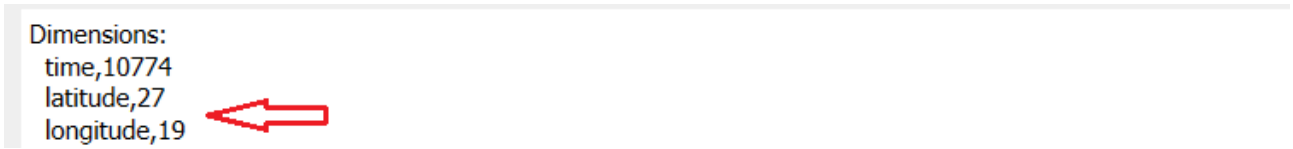
To create a mask in Xraster, a **vector polygon layer** in a geographic coordinate system is most commonly used. A unique identifier is assigned to each relevant area in a selected attribute field.

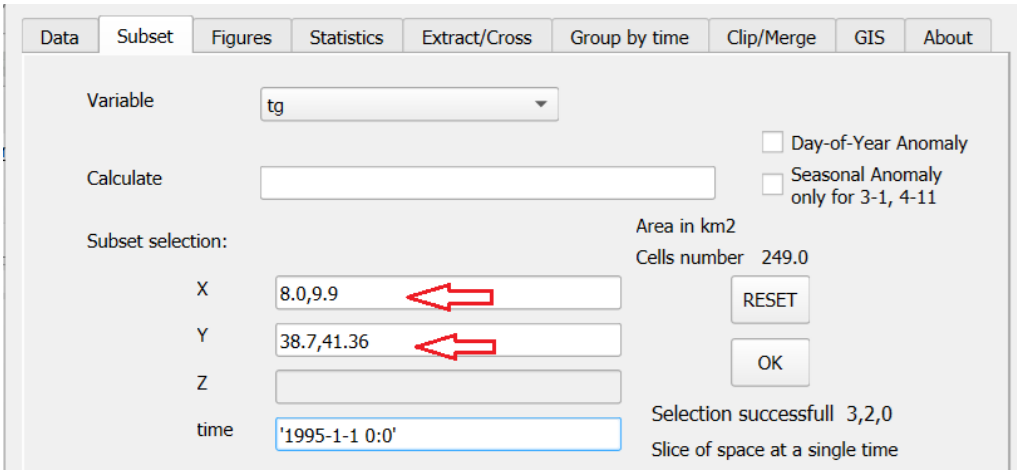
The **land layer** was downloaded from Natural Earth and the value **1** was then assigned to the **cod** field.



The first step is to prepare a new file from the **.nc dataset** containing a single layer with the same number of rows and columns as the original raster. To do this, we open the dataset and note the number of rows and columns of the raster.

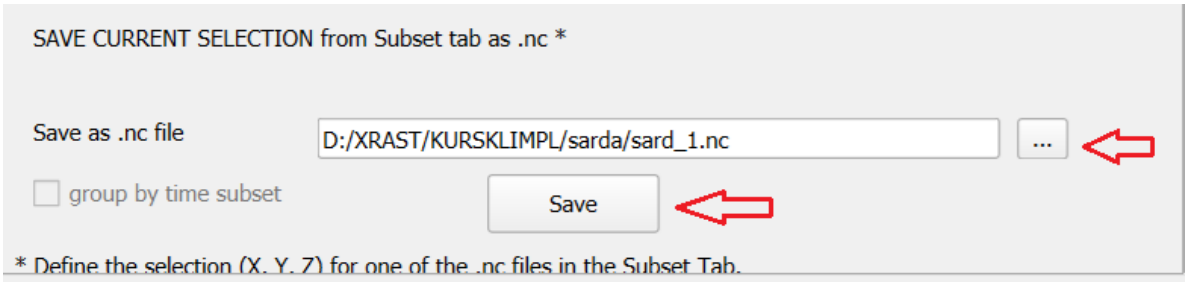
For the previously created dataset, there are **27 rows and 19 columns**.



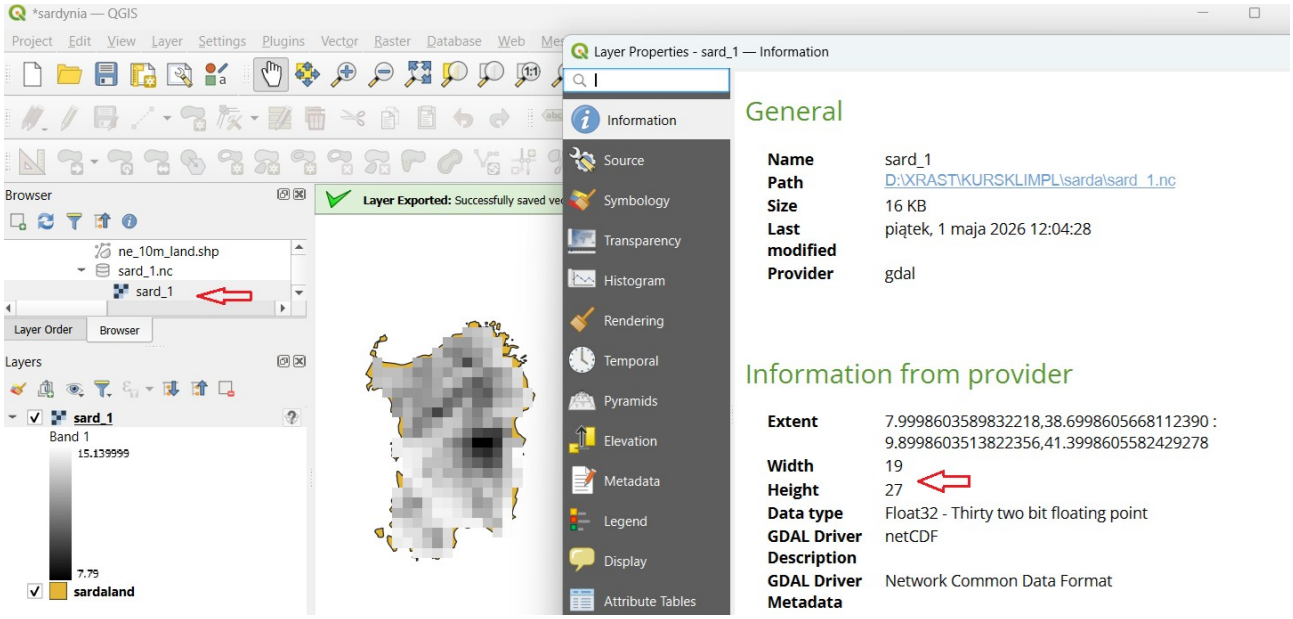


We create a subset of our data for a single day. We adjust the latitude and longitude ranges by approximately half the size of a grid cell to obtain exactly the same number of rows and columns.

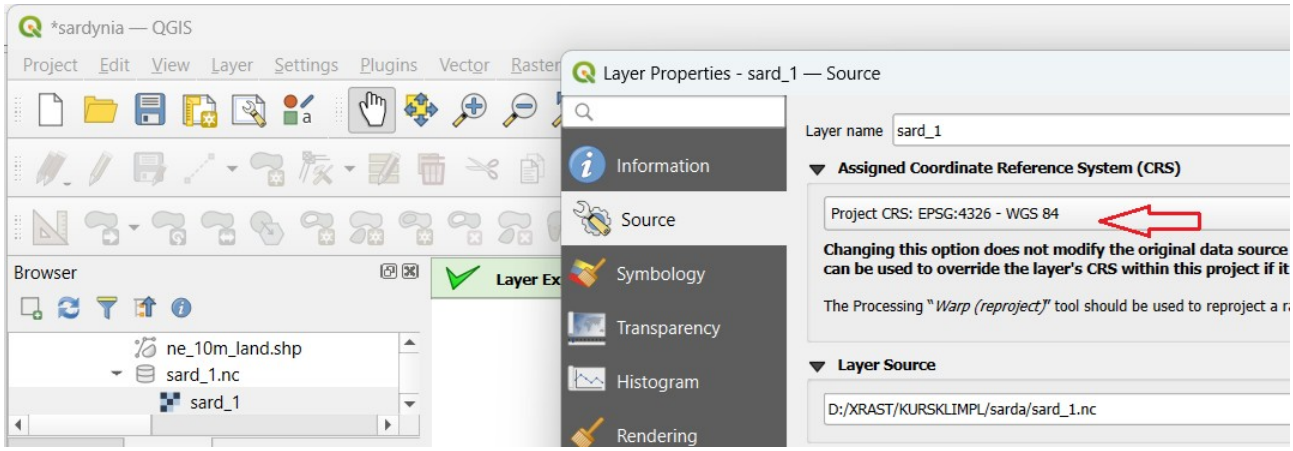
Next, we export this subset as a new **.nc dataset** using the **Clip/Merge** tab.



Opening the newly created dataset allows us to load the raster and verify the number of rows and columns.

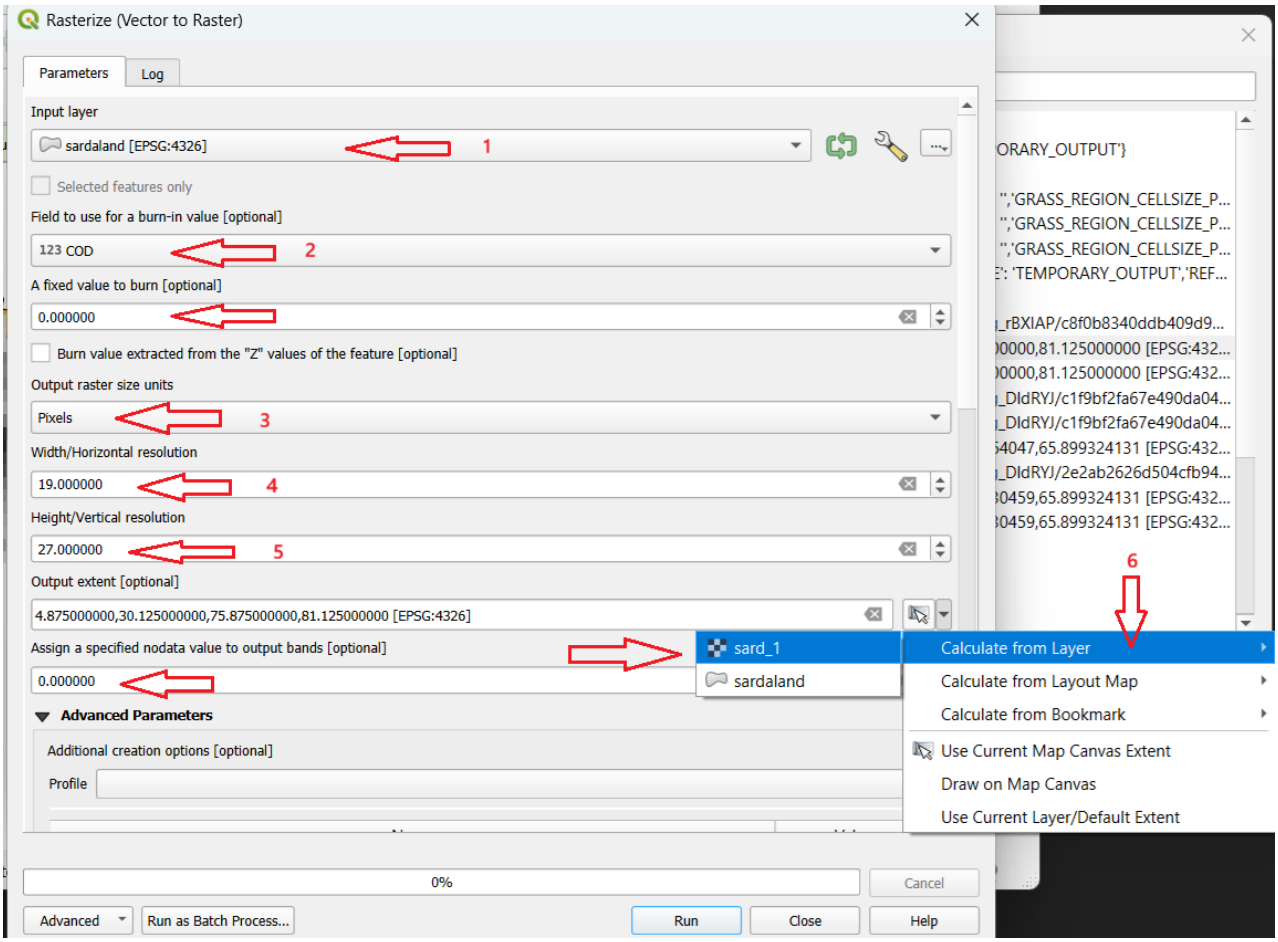


Dodatkowo należy przypisać warstwie układ współrzędnych,



This layer will be used to define the dimensions of the mask.

The mask is created in QGIS using two tools: **Rasterize** and **Fill NoData Cells**.

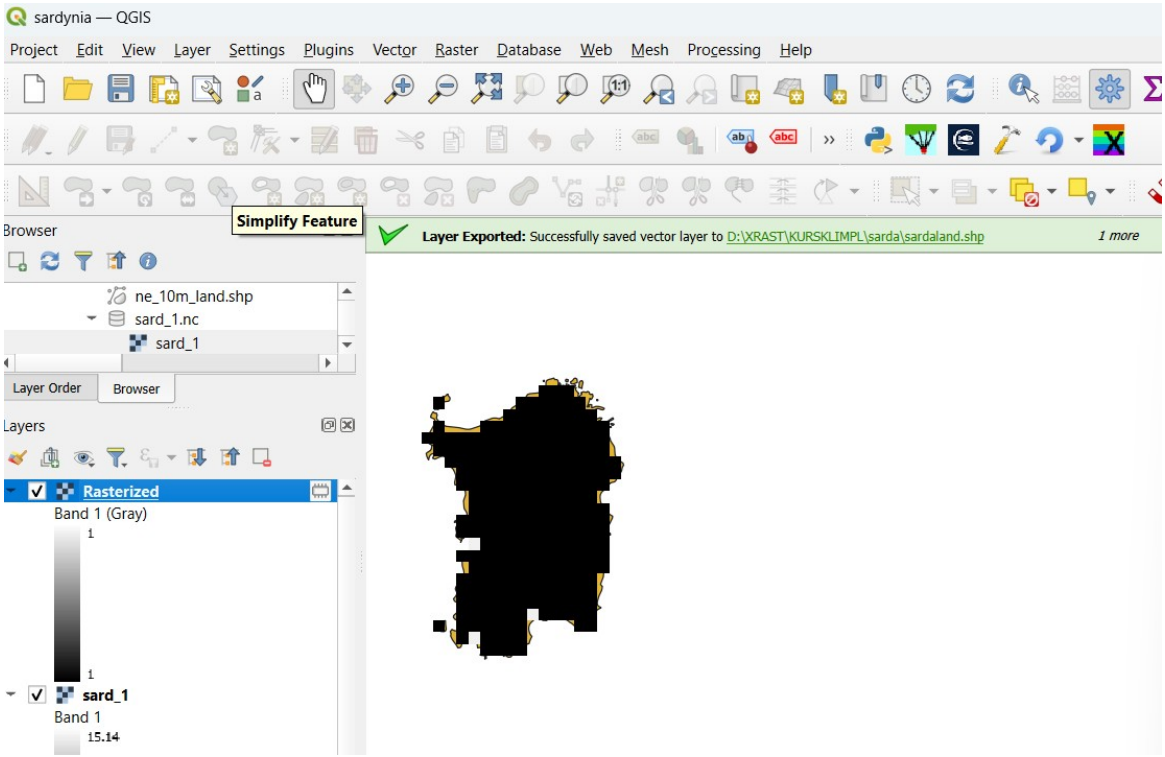


Open the **Rasterize** tool. Select the vector polygon layer **(1)**, select the field containing the area identifiers **(2)**, and set **Output raster size units** to **Pixels (3)**.

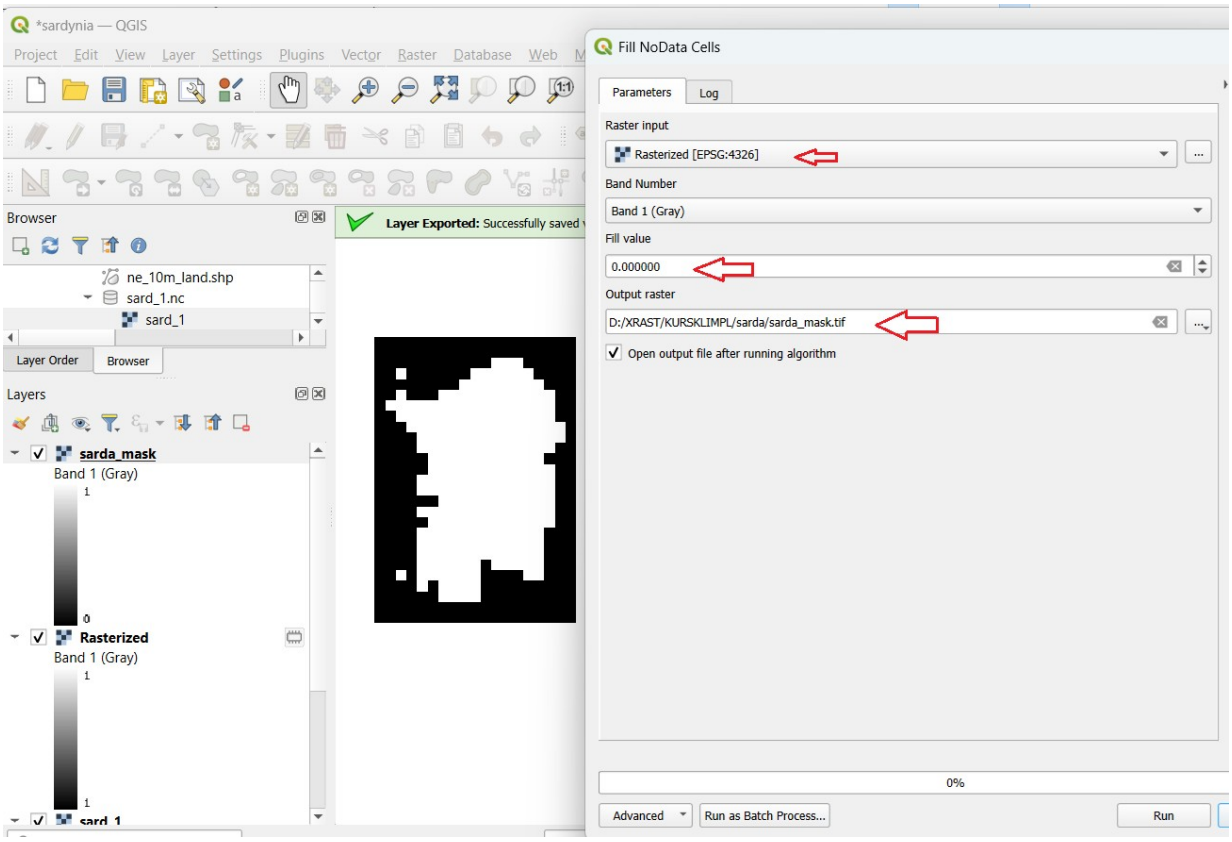
Enter the horizontal and vertical resolutions **(4, 5)**. Set the **Output Extent** using the raster layer from the **.nc** dataset **(6)**.

Click **Run**.

A temporary layer named **Rasterized**, resulting from the rasterization process, is created.



The next step is to fill the **NoData** cells with a value of **0** using the **Fill NoData Cells** tool.



Select the rasterized layer (**Rasterized**), enter **0** as the **Fill value**, and specify the name of the output mask.

The mask can now be loaded into **Xraster**.

